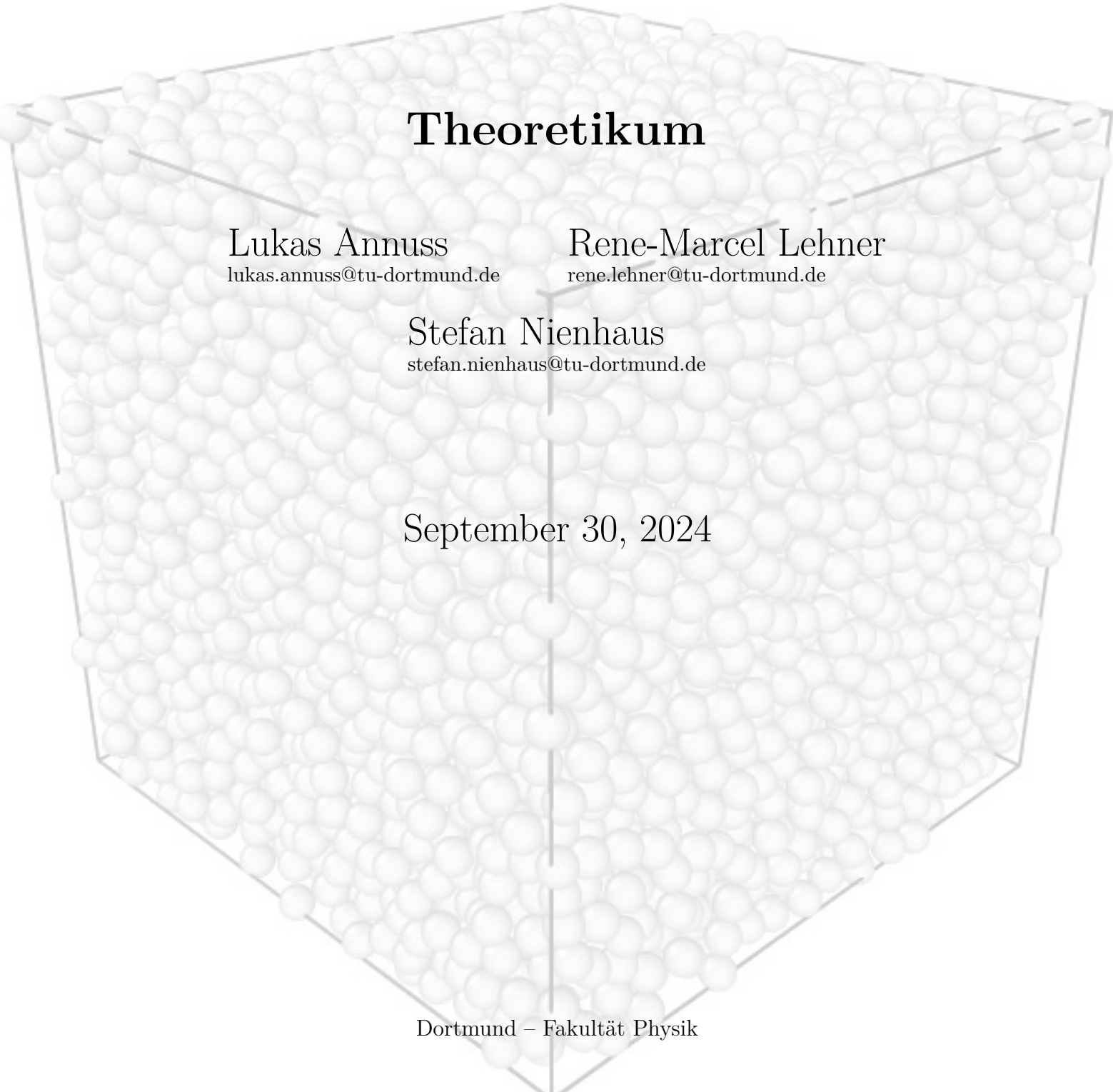




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1 Abstract [L. Annuss] with support from [R.-M. Lehner]

The hard sphere system displays, despite its simplicity, a wide amount of behaviors. For this reason, it forms the basis for more complex systems like polymers. In this study we will focus our exploration of this behavior on the formation of crystals and the sedimentation process in three dimensions and the Kosterlitz-Thouless phase transition in two dimensions. To simulate the hard disk system, we will first implement a fast non-reversible Markov-Chain algorithm that breaks detailed balance and explain why this rejection-free algorithm is especially suited to sample the Boltzmann distribution of the hard sphere system. Then we will explain how one can calculate observables like pressure and the pair correlation function $g(r)$ from an algorithm that disregards the physical dynamics of the hard sphere system. Following this, we will use these observables to compare our own implementation of the event chain algorithm with theoretical prediction. To characterize the crystals that are formed by the entropy driven crystallization process in dense systems, we use the Q_6 , q_4 , q_6 and q_6q_6 order parameters and the Minkowski tensors. Here Q_6 will help in determining the degree of order of the whole system. q_6q_6 and the Minkowski structure metric will be used to assign a phase or crystal type to the individual particles. We show that the hard sphere system forms an FCC crystal which is interspersed with HCP lattice layers.

To investigate the Kosterlitz-Thouless phase transition, we explore the equilibrium equation of state near the transition point and use the positional and orientational order functions to classify the phases of the two-dimensional system. We find a continuous phase transition between the liquid phase and an intermediate hexatic phase and a second Kosterlitz-Thouless transition between the hexatic and solid phase.

Lastly, we extended the event-chain Monte Carlo algorithm to simulate sedimentation by introducing an external gravitational potential and fixed boundaries. Our simulations revealed the formation of a solid phase in the lower layers and a phase transition occurring in the middle layers due to gravity-induced density gradients. By measuring the height-dependent packing fraction and pressure, we observed reasonable agreement with the Carnahan-Starling equation of state. Structural analysis identified the development of FCC and HCP crystal structures within the sedimented layers. These findings highlight the complex behaviors that arise in hard-sphere systems under gravity, demonstrating the utility of event-chain algorithms in exploring sedimentation dynamics and phase transitions.

This study highlights the complexity of the hard sphere system and expands the knowledge of its phase transitions and crystallization, but it also demonstrates the applicability of an event chain driven approach to explore physical systems.

2 Basics of simulating hard spheres [S. Nienhaus]

In this chapter the system of hard spheres is introduced first. The start- and boundary conditions are discussed and how such a system develops over time. To measure the development the common parameters being the pressure and pair correlation are explained. Basic Monte Carlo simulations are expounded as preparation for the event-chain algorithm. The straight event-chain algorithm is discussed in detail by explaining how a single event is calculated and by exploring how its features can simplify the pressure measurement. Furthermore it is discussed why the the event-chain algorithm is a valid and fast Monte Carlo algorithm. The last section will display measurements of pressure and pair correlation using an implementation of the event-chain algorithm made by the authors of this report.

2.1 Hard Spheres [S. Nienhaus]

A hard sphere is characterized by a diameter σ and a potential

$$v(r) = \begin{cases} \infty & r < \sigma \\ 0 & r \geq \sigma \end{cases} \quad (1)$$

meaning that the potential here is simply given by the fact that spheres cannot overlap. In this study the simulation volume is a cube with periodic boundary conditions. The rest of the system, apart from the individual sphere positions, is now defined by the packing fraction $\eta = \frac{V_{\text{spheres}}}{V_{\text{volume}}}$ on which the phase behavior of the system depends. The independence of the phase behavior on the temperature can be seen when inspecting the boltzmann factor

$$e^{-\beta v(r)} = \begin{cases} 0 & r < \sigma \\ 1 & r \geq \sigma \end{cases} \quad (2)$$

which itself is independent of the temperature due to the form of Equation 1 of the potential.

To actually determine the equation of state of the system a virial expansion

$$p\beta = \sum_{l=1}^{\infty} B_l(T) \rho^l$$

with $\beta = \frac{1}{k_B T}$ and $\rho = \frac{N}{V_{\text{volume}}}$ the particle density of the system is used. The general approaches for calculating the coefficients B_l are rather complicated. Further details can be found in [1, pp. 19–31]. For hard spheres the virial expansion can be rewritten using the packing fraction η as

$$\frac{p\beta}{\rho} = 1 + \sum_{l=2}^{\infty} \underbrace{\frac{B_l}{v_c^{l-1}}}_{=B_{l-1}} \eta^{l-1}.$$

The coefficients B_i have been calculated and it was found that they approximately follow

$$B_i \approx a_1 i^2 + a_2 i + a_3 = i^2 + 3i$$

resulting in

$$\frac{p\beta}{\rho} = 1 + \sum_{i=1}^{\infty} (i^2 + 3i)\eta^i = \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3} \quad (3)$$

which is the Carnahan-Starling equation of state.

While the Carnahan-Starling equation is an approximation, an exact equation is given by

$$\frac{p\beta}{\rho} = 1 + 4\eta g(\sigma+). \quad (4)$$

$g(\sigma+)$ is the contact value of the pair correlation function. The contact distance $r = \sigma+$ is the minimum distance two spheres can have without overlapping. The pair correlation function describes the probability of finding a particle at a distance r from another particle. Therefore measuring it involves determining distances between all particles and taking care of periodic boundary conditions. Since this procedure is expensive, a better way for calculating the pressure will be presented in Equation 11. Binning these distances while preventing double counts results in the binned radial pair correlation function $g(r)$. Therefore $g(r)$ describes the average number of particles that is expected inside a bin $[r + dr]$ for a particle at $r = 0$. $g(r)$ can be calculated via

$$g(r) = \frac{dn(r)}{\rho N dV},$$

where dV is the volume element of the system and $dn(r)$ is the histogram of pair distances with a bin size of dr . [2]

2.2 Monte Carlo Simulations [S. Nienhaus]

In contrast to molecular dynamics simulations, Monte Carlo Simulations do not aim at simulating all mechanical processes in detail but instead they return thermodynamic means of the system. These means are constructed by averaging over the boltzmann distributed states. Therefore the goal of a Monte Carlo Simulation is to sample the phase space of the system according to the boltzmann distribution

$$p_i = \frac{1}{Z} e^{-\beta E_i} \quad [2]. \quad (5)$$

This is done by introducing at least one of the following conditions to ensure that the system is converging against the required equilibrium distribution of states p_i . In this case p_i can be a general probability distribution. To ensure convergence the probability currents or probability flows need to be chosen such that the probability distribution of states becomes stationary at equilibrium for the desired distribution p_i .

The weaker of the two conditions is global balance and states that the sum of all probability currents into state i is equal to the sum of all probability currents out of state i meaning $\sum_j \mathcal{F}_{ji} = \sum_k \mathcal{F}_{ik}$ with $\mathcal{F}_{ij} = p_i M_{ij}$ resulting in

$$\sum_i p_i M_{ij} = p_j. \quad (6)$$

M_{ij} is the probability to transition from state i to state j . Note that this condition still allows circular probability currents, but these do not break the stationarity of the distribution of states. Global balance always needs to be fulfilled because otherwise the system will not fulfill the requirement to converge against the target distribution. The more strict condition is detailed balance which states that the probability current from state i to state j is equal to the probability current from state j to state i leading to $F_{ij} = F_{ji}$ or

$$p_i M_{ij} = p_j M_{ji}. \quad (7)$$

This way the target distribution is preserved after every transition. Circular probability currents are not allowed with detailed balance. This condition is not strictly necessary to construct a valid Monte Carlo method but global balance is.[3]

Now the task is to construct transition probabilities M_{ij} that fulfill the previous conditions. A possible solution is given by

$$M_{ij} = V_{ij} A_{ij} \quad \text{with} \quad A_{ij} = \min \left(1, \frac{p_j}{p_i} \right) \quad \text{for } i \neq j. \quad (8)$$

V_{ij} is the probability to consider the transition $i \rightarrow j$ in the first place and needs to fulfill $V_{ij} = V_{ji}$. The probability to accept this suggested transition is given by A_{ij} . This is called the Metropolis algorithm and satisfies detailed balance. For the case of an actual Monte Carlo simulation with the boltzmann distribution as target this translates to

$$A_{ij} = \min \left(1, e^{-\beta(E_j - E_i)} \right). \quad (9)$$

For hard spheres due to Equation 2 this acceptance probability becomes

$$A_{ij} = \begin{cases} 0 & r < \sigma \\ 1 & r \geq \sigma \end{cases} \quad (10)$$

meaning that for a simple local Monte Carlo only the two situations, either accept move because it creates no overlap or reject move because it does create an overlap need to be considered. [2]

2.3 Event-Chain Algorithm [S. Nienhaus]with support from [L. Annuss]

The event-chain algorithm (ec-algorithm) is a Monte Carlo method and as such it is fundamentally different from an molecular dynamics simulation as it does not use physical movement steps considering momentum and collision angle of spheres. It is also non-deterministic as it uses random sampling. Therefore many physical features of a hard sphere system cannot be investigated using the ec-algorithm since the detailed dynamics are not simulated. Why is it used then? The most important reason is that it is much faster than molecular dynamics simulations. This becomes especially advantageous for processes like crystallization that can take a long time to occur. In many studies thermodynamic means and not the precise dynamics of a system are of interest. Although the concept of an acceptance probability does not apply in an event chain, the fundamental

requirement that a move must merely avoid creating overlaps remains, which simplifies the construction of events for the ec-algorithm.

An event-chain can be constructed out of events and chains. An event describes the movement of a single sphere which also includes the collision detection. On collision the previously moving sphere stops and the other sphere starts moving. A chain usually consists out of multiple events and is started by randomly choosing a particle whose direction is also chosen randomly. Individual events are performed until the combined movement of all spheres in the chain reaches a predefined chain-length l . There are two types of event-chain simulations, the straight event-chain and the reflected event-chain. For the reflected event-chain a sphere changes the chain's direction on collision with respect to the symmetry axis of the collision. For the straight event-chain this change of direction does not occur within one chain and all spheres continue their movement in the same direction as the sphere which started the chain as it is depicted in Figure 1. In this work only the straight event-chain is used.

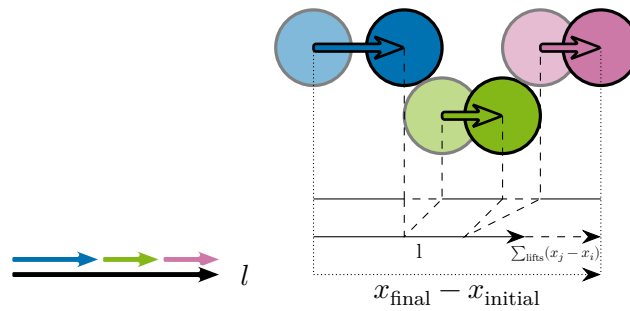


Figure 1: Depiction of a straight event-chain consisting out of three events. For each collision there is an additional displacement created within the chain that does not count to the chain-length l . This additional displacement can be used to calculate the pressure of the system. (adapted from [3] and [4])

After one chain has ended a new chain is started with a new randomly chosen sphere. Apart from a chain there exists a sweep which consist out of multiple chains. During one sweep each sphere should be moved at least once on average. Sweeps are not strictly necessary to perform an ec-algorithm but become unavoidable when dealing with processes that occur over some timescale. Sweeps are the only way to define such a timescale for the ec-algorithm and to compare the results of different simulations.

The outline of one individual collision event is shown in Figure 2. $\vec{r}_1$ and $\vec{r}_2$ are the positions of the two spheres before the collision. $\vec{v}$ is the direction of the chain and has a length corresponding to the chain-length that is still left. t is the collision parameter that might has to be calculated and is in the range of $[0, 1]$.

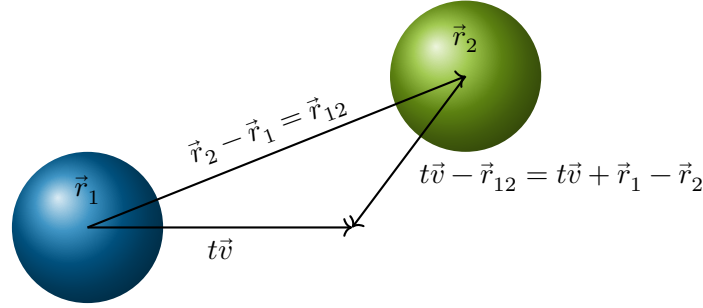


Figure 2: The Geometry of a collision event between two spheres is shown. From the distance vector $\vec{r}_{12}$ at the beginning and the distance vector $t\vec{v} - \vec{r}_{12}$ of spheres on impact the possible movement distance $t\vec{v}$ can be calculated.

Before directly calculating the collision parameters it is recommended to check whether a collision is possible first. This helps in reducing computation times and causes the resulting collision parameters to not needing further treatment. In order to achieve this the parameter $b = \vec{v} \cdot \vec{r}_{12}$ is defined. If $b < 0$ the spheres are moving away from each other and a collision is impossible. The condition for a collision is, as can be directly seen in Figure 2,

$$\begin{aligned}
 \sigma &= |t\vec{v} - \vec{r}_{12}| \\
 \Leftrightarrow \sigma^2 &= (t\vec{v} - \vec{r}_{12})^2 \\
 \Leftrightarrow \sigma^2 &= t^2\vec{v}^2 - 2t\underbrace{\vec{v} \cdot \vec{r}_{12}}_{=b} + \vec{r}_{12}^2 \\
 \Leftrightarrow 0 &= t^2\vec{v}^2 - 2tb + \vec{r}_{12}^2 - \sigma^2 \\
 \Leftrightarrow t &= \frac{b \pm \sqrt{b^2 - \vec{v}^2(\vec{r}_{12}^2 - \sigma^2)}}{\vec{v}^2}.
 \end{aligned}$$

There are two possible collision points but the one with the smaller parameter t (solution with minus) appears first in the simulation making the other solution with plus irrelevant. With that formula a second condition for a collision can be found as the value inside the square root has to be positive. Therefore the new condition is $b^2 - \vec{v}^2(\vec{r}_{12}^2 - \sigma^2) > 0$.

In a real simulation there are more than two spheres and only the sphere with the lowest collision parameter t collides with the active sphere. In a very basic implementation every sphere has to be checked for a collision to find the one with the lowest t . For large numbers of spheres this becomes impractical and methods like neighbor lists need to be used.

Before a ec-simulation can be started an initial particle configuration needs to be constructed. In many cases a random configuration is required which is usually very difficult to construct for high packing fractions. Randomly choosing positions at dense packing fractions will result in many overlaps. The ec-algorithm can solve this issue by starting chains at overlapping spheres until these overlaps are resolved. This is called ec-rattling and can help in constructing dense random initial configurations.

For event-chain simulations the pressure measurement can be simplified. The formula for measuring the pressure is given by

$$\frac{\beta p}{\rho} = \left\langle \frac{x_{\text{final}} - x_{\text{initial}}}{l} \right\rangle_{\text{ECs}}. \quad (11)$$

l is the chainlength, x_{final} and x_{initial} are the positions of the final and initial sphere of the chain. The average runs over several chains. This technique uses the additional displacement length that is accumulated in a chain for each collision event as shown in figure 1. For a more detailed explanation see [4].

Why does the ec-algorithm work as an extraordinarily fast MC-algorithm? First it should be clear why the ec-algorithm is working at all as a valid MC-method. To understand this the local version of that algorithm is introduced. Here no chain is performed and for each event a new sphere and direction is chosen. Next the move in the proposed direction can either be accepted or rejected. It is obvious that the detailed balance condition is satisfied when all directions are chosen with equal probability. However, it should be verified that the probability of reaching a certain configuration a is equal to the Boltzmann weight for hard spheres, which should be 1 to satisfy global balance. This can be used later on to proof the validity of the ec-algorithm. For only having to deal with two directions „–“ backward and „+“ forward the 1D case is considered. Furthermore small discrete moves of certain length instead of the refined calculations above are used in the local MC.

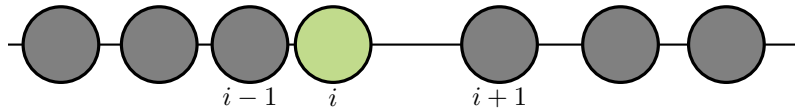


Figure 3: Sketch of a 1D hard spheres system.

The calculation for this probability flow into configuration a is given by

$$\mathcal{P}_a = \frac{1}{2N} \sum_i (\mathcal{A}_i^+ + \mathcal{R}_i^- + \mathcal{A}_i^- + \mathcal{R}_i^+). \quad (12)$$

$\mathcal{A}_i$ is the probability for a sphere i to be offered a move in the direction $+/-$ and accept it to reach configuration a . $\mathcal{R}_i$ is the probability for a sphere i to be offered a move in the direction $+/-$ and reject it to reach configuration a . N is the number of spheres in the system. The summation over all spheres is necessary because in local MC some random sphere needs to be chosen at the beginning of an event. From these definitions the two conditions

1. $\mathcal{A}_i^+ + \mathcal{R}_i^- = 1$
2. $\mathcal{A}_i^- + \mathcal{R}_i^+ = 1$

arise. These justify the result in Equation 12. To understand this, consider Case 1 and assume that a move in the forward direction has been accepted, so $\mathcal{A}_i^+ = 1$. In this situation, it is impossible for a move leading to configuration a in the backward direction

to have been rejected, as there was sufficient space to the left of particle i for the forward move to occur. Similar reasoning applies to all other cases.

For the event chain the same result for the flow can be shown with help of Figure 4. The formula for the flow is

$$\mathcal{F}(t, i)_a = \mathcal{A}_i + \mathcal{R}_{i-1} = 1. \quad (13)$$

For the event-chain only one particle is able to move at time and is determined through collisions removing the sum over all particles from the flow equation. Additionally the cases for other directions like forward and backward in the 1D example for the local MC can be discarded as the chain is always moving in the same direction. A new variable has been introduced the lifting variable i . This variable redirects the rejected probability flow to a change of the lifting variable. In the case of hard spheres this lifting variable is simply the particle ID of the active sphere. Equation 13 always needs to be equal to 1 as can be seen when exploring all cases. When a move from i is accepted ($\mathcal{A}_i = 1$) to reach configuration a it cannot be the case that a move from another sphere has been rejected ($\mathcal{R}_{i-1} = 0$) before as it would not have been the active particle in the previous event. This can be seen in Figure 4 for the frames (a, green) and (b, green). The active spheres are marked by arrows showing the proposed moves. To the left of the active green sphere there has been enough space for the move of green to be accepted. In turn this means that a rejection for red and a following lifting move from red to green could not have happened. For (b, red) the lifting move has happened leading to the lifting probability flow. This configuration could have only happened through the lifting move. A flow coming from configuration (a, red) is not possible as an accepted move from red would mean that red would have been in an overlapping configuration before which is forbidden. The probability flow in Figure 4 continues to (c, red) where the move of the now active red particle has been accepted. A probability flow into the invalid configuration (c, green) is not possible as an acceptance of the green move is prevented due to overlaps ensuring that the probability flow goes only into valid configurations. Now it is understood why the ec-algorithm is a valid MC-algorithm as all flows are equal to 1 and therefore global balance is satisfied.

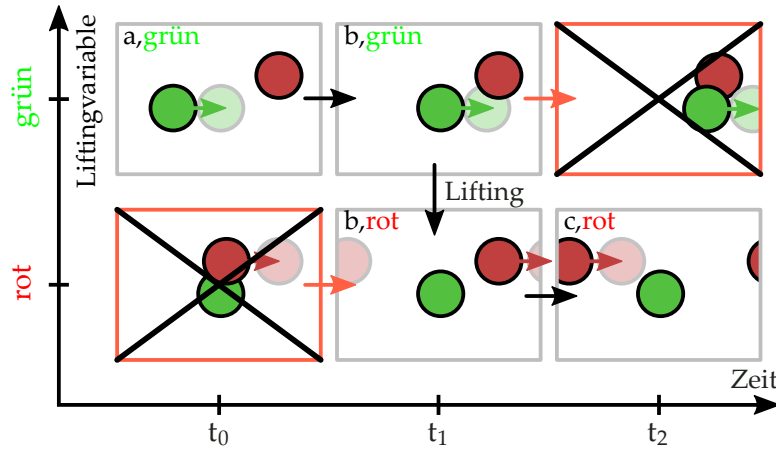


Figure 4: Explanation of the lifting formalism to create rejection free ec-moves to maximize global balance. Green and red arrows mark the active sphere. [3]

Why is it exceptionally fast ? The aim of every MC-simulation is to reach equilibrium as quickly as possible. By using the event-chain algorithm with lifting the detailed balance is partly broken, at least within one chain. If all directions are sampled for several chains detailed balance is satisfied over several chains because reverse moves can be proposed with the same probability again. Some special ec-algorithms that do not sample from all directions but only along coordinate axes, called xy-algorithms, do not even satisfy detailed balance over several chains [3]. However within one chain only global balance is satisfied. To be precise, due to the lifting formalism global balance is maximized, meaning that the probability flow goes into one direction only as shown in Figure 5. This means that the algorithm converges as fast as possible to equilibrium. Furthermore by the lifting formalism as it is used above when using the remaining chain-length true rejections are avoided.

This leads to the discussion of which chain length l is appropriate. On the one hand the chain has to be long enough such that it can be made use of the lifting feature of the algorithm. If l is too short no rejection would have been taken place anyway and the ec-algorithm would be similar to a local Monte Carlo. On the other hand if l is too large all spheres could be displaced in a single event chain, leading to a translation of the whole system. In this case the new system would be highly correlated with the system before. The Boltzmann distribution would not be analyzed efficiently with different independent samples.

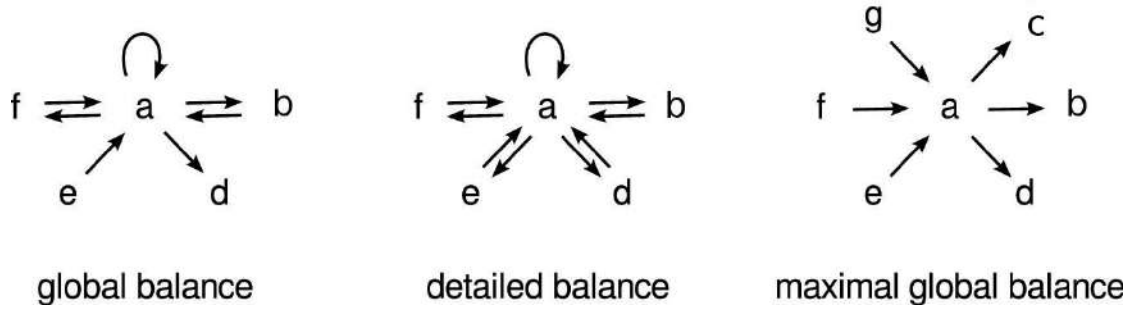


Figure 5: Possible probability flows for MC-algorithms according to the three conditions detailed balance, global balance and maximum global balance. [5]

2.4 Measuring $g(r)$ and $\beta p/\rho$ using the ec-algorithm [S. Nienhaus]

In this section the pair correlation function and the equation of state are determined using the ec-algorithm implemented by the authors of this work. The results are compared to the literature and to the formulas given in subsection 2.1. For the different simulations only η is varied. The other simulation parameters are given in Table 1. The pair correlation as shown in Figure 6 can be used to differentiate between gaseous, fluid or crystalline systems. Additionally the pair correlation from the crystallized simulation state Figure 17c is shown.

Table 1

L	20
σ	1
l	20

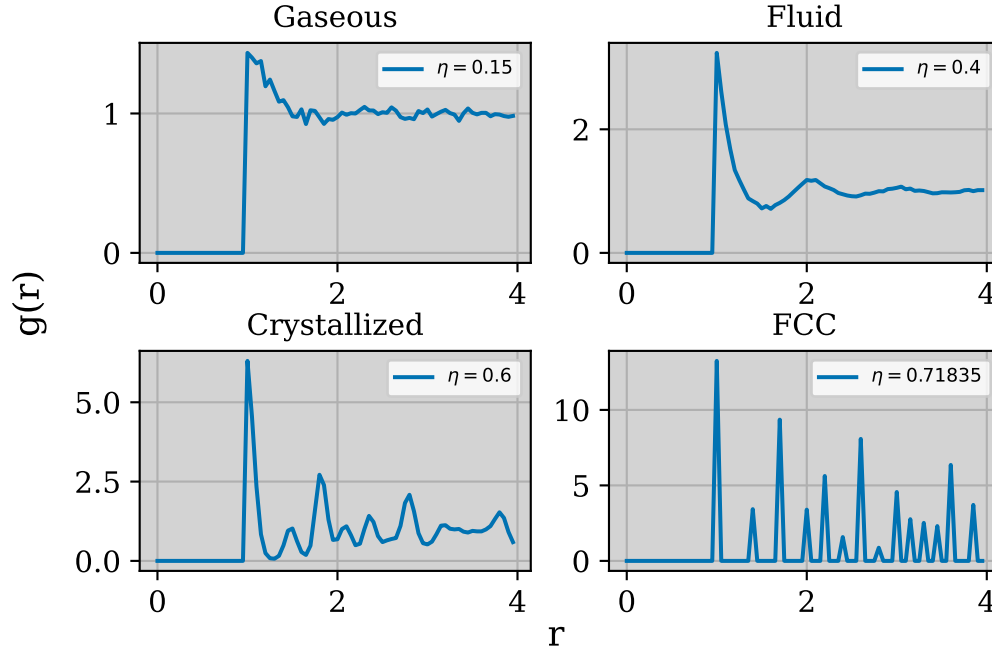


Figure 6: Pair Correlation for a gaseous phase ($\eta = 0.15$), a fluid phase ($\eta = 0.4$) and a fcc crystal ($\eta = 0.71835$). The crystallized simulation state has been generated at $\eta = 0.6$ with an external implementation of the event-chain algorithm [6].

For a gaseous system there exists only one maximum at $r = \sigma = 1$ which is the most probable distance between two spheres. For the fluid phase there are many maxima with decreasing height due to the short range order of the fluid. The short range order can be described by shells leading to the characteristic maxima of the fluid phase. For a crystal, in this case an FCC crystal, there are again infinitely many sharp maxima due to the long range crystalline order. Therefore when the system is not in a crystalline phase the limit $\lim_{r \rightarrow \infty} g(r) \approx 1$ should hold true. This occurs because two particles that are separated by a large distance should not be correlated with each other outside the crystalline phase. [2] The example from a crystallized simulation state contains large FCC regions separated by HCP strips and small fluid patches. Therefore the FCC peaks are visible but sit on top of a small HCP and fluid background.

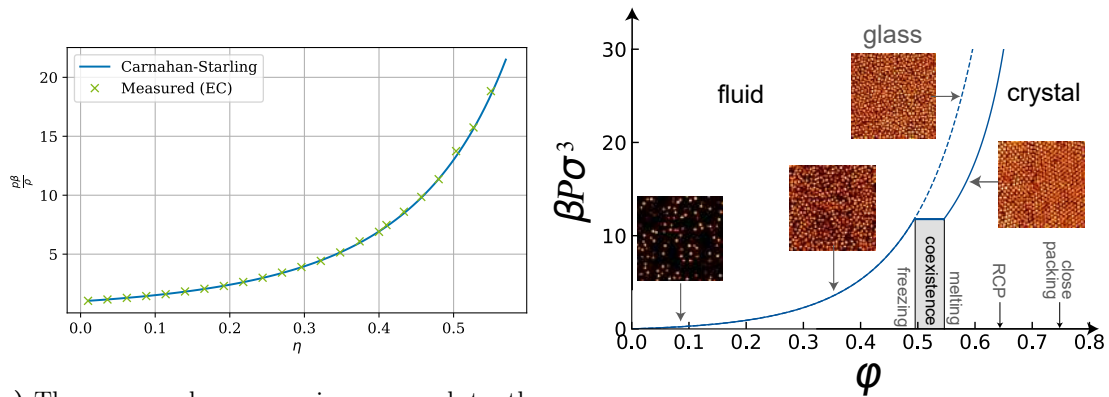
From the pair correlation the contact values $\sigma + (\eta = 0.15) = 1.43462$ and $\sigma + (\eta = 0.4) = 3.237$ for gas and fluid can be extracted. Using these contact values and the pressure measurement with Equation 11 the pressure can be calculated to show that these methods work and the implemented ec-algorithm is returning the expected results.

Table 2: Pressure determined in three different ways, using Equation 11, Equation 3 and Equation 4 for the gaseous and fluid phase. Note that for the FCC crystal no simulation is performed and no ec-pressure therefore exists. The crystallized configuration has been generated with polyeventchain [6] and therefore is ignored here as well.

	$\frac{\beta p}{\rho_{ec}}$	$\frac{\beta p}{\rho_g}$	$\frac{\beta p}{\rho_{Carnahan-Starling}}$
$\eta = 0.15$	1.899	1.861	1.904
$\eta = 0.2$	6.949	6.179	6.926

The values obtained with the contact values differ a lot from the Carnahan-Starling result due to the limited particle number and therefore limited statistics.

The pressure can be plotted against the packing fraction and is calculated using Equation 11. It is again compared to the Carnahan-Starling equation in Figure 7a. The pressure curve from the literature is given in Figure 7b. When comparing both plots it becomes clear that only the metastable glass region has been explored in Figure 7a. This happens because of the slow implementation preventing the simulation from leaving that metastable glass phase. In a more refined simulation that runs for longer the hard spheres will undergo crystallization processes which will be explained in section 4 in detail. The result is a plateau of the equation of state in the so called coexistence region. After that the pressure rises again.



(a) The measured pressure is compared to the Carnahan-Starling approximation. (b) Literature result for the equation of state depending on the packing fraction. [7]

3 Advanced Simulation Techniques [R.-M. Lehner]

While a basic hard-sphere simulation can be implemented with a relatively small code-base—requiring only careful management of entities such as the simulation box and spheres—two significant challenges must be addressed: collision detection and the efficient determination of neighboring particles.

Although the method for detecting collisions has been covered previously, efficiently identifying potential collision partners remains crucial for the performance of the simulation. In large systems, checking every particle against all others is computationally infeasible due to the $\mathcal{O}(N^2)$ scaling. Therefore, efficient methods are required to reduce the computational cost.

In this section, we explore advanced techniques for efficiently determining potential collision partners, briefly discussing neighbor lists and grid-based methods.

3.1 Neighbor Lists

Detecting collisions requires comparing the positions and distances to other particles. naïvely performing this for every particle pair leads to computational costs that scale as $\mathcal{O}(N^2)$, which is impractical for large simulations. Intuitively, only nearby particles—those within collision range and in the relevant direction—need to be considered. Implementing this intuitional method requires efficient data structures to manage particle neighborhoods. A common approach is the use of neighbor lists, such as Verlet lists [8]. In this method, each particle maintains a list of nearby particles within a specified cutoff radius. By carefully choosing this cutoff radius, neighbor lists do not need to be updated every iteration, reducing computational overhead.

Introducing such lists significantly reduces the number of candidate pairs for collision checks. However, updating these lists can still be computationally expensive, as it may require scanning over all particles regularly to identify neighbors.

3.2 Introducing a Grid

To further minimize computational costs, we introduce a grid-based method that reduces the need for frequent updates and efficiently identifies the minimal environment where collision partners may exist.

Instead of maintaining neighbor lists for each sphere, we partition the simulation volume into a grid of cells—small cubes or cuboids—and track which spheres are located in each cell.

Care must be taken to ensure that the cells fit evenly into the simulation volume to avoid undesirable edge effects. We choose the sphere diameter σ as the lower limit for the cell side length. The actual side length is determined by stretching σ to fit an integer number of cells into the simulation volume. This process is illustrated in Figure 9. The stretched side length a is determined by equations (14).

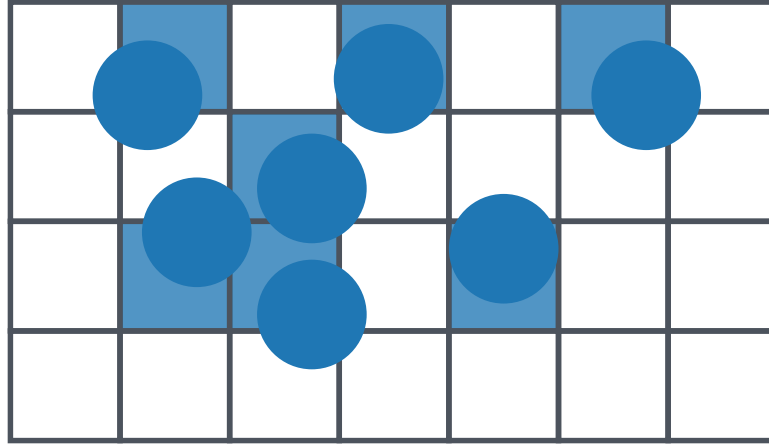


Figure 8: Grid partitioning of the simulation volume into cells. Blue cells contain spheres. Overlaps into multiple cells are not relevant; only the center point of each sphere is considered.

$$\begin{aligned} n &= \left\lfloor \frac{L}{\sigma} \right\rfloor \\ a &= \frac{L}{n} \end{aligned} \tag{14}$$

The grid updates dynamically whenever a sphere moves and crosses a cell boundary. By introducing the grid as shown in Figure 8, we effectively create an external neighbor management system. Each sphere knows which cell it occupies and can directly query adjacent cells to find nearby spheres. This method simplifies neighbor determination by localizing interactions. However, the actual process of identifying neighbors within and across cells still needs to be addressed.

3.2.1 Motivation for the Cell Size

The fundamental idea is to choose cells as small as possible to explore only the necessary neighborhood. A sphere can always be located in exactly one cell (despite its expansion) because only the center point of the sphere is considered. However, as illustrated in Figure 10, collision partners from adjacent cells can still be relevant. These edge effects occur regardless of the cell size.

A cell size of σ is the smallest dimension that guarantees all potential collision partners are identified when all surrounding cells are considered. Again, this is illustrated in Figure 10, where the **blue cell** contains the particle being considered. The **orange cells** are a margin around the blue cell that can contain potential collision partners. A cell size of equal or greater than σ ensures that spheres in the **gray cells** are not reachable from within the blue cell.

Of course, when the sphere moves to a new cell, new neighboring cells must be considered.

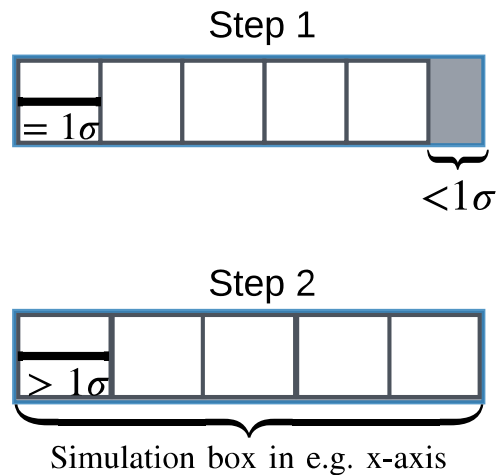


Figure 9: Example of building the grid in a pseudo 2D simulation box. The cell side length is determined by stretching σ to fit an integer number of cells into the simulation volume.

Step 1: Fill the grid with as many n cells of side length σ as possible.

Step 2: Stretch σ to fit these n cells into the entire simulation volume. Repeat this process for all dimensions.

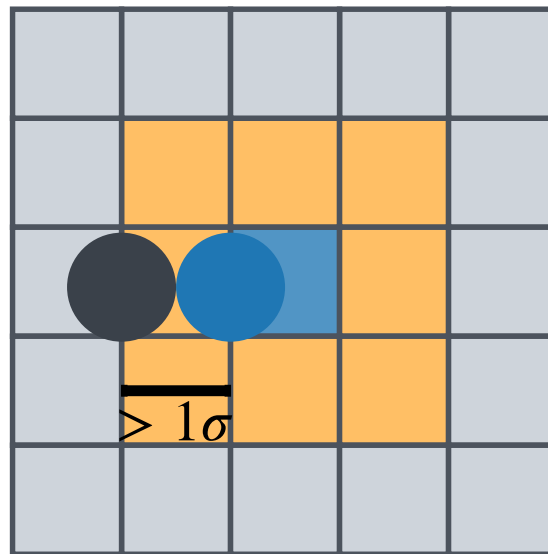


Figure 10: Motivation for choosing the minimal cell size: minimizing the distance to unreachable spheres. This is the closest distance a sphere from the gray cells can be from the blue cell.

Blue cell is assigned to the particle being considered.

Orange cells are those cells that can contain potential collision partners reachable from within the blue cell.

Gray cells contain spheres that can no longer be reached from the blue cell.

3.2.2 Naïve Approach

A simple method to identify potential collision partners is to interpret the possible flight length (displacement length) of a particle as a radius and determine the neighbors within this radius. An iterative approach is sensible here, where the cell-shell radius is increased by 1 in each iteration after a missed collision. However, in a three-dimensional space, this means the number of cells considered per iteration grows by $\mathcal{O}(n^3)$.

3.3 Ray Tracing

The new method follows the idea of considering only the cells that are actually traversed. The direction vector is treated as a ray that pierces cells along its path. This task is solved using a ray tracing algorithm 1. The algorithm is very fast and can be executed iteratively, allowing cells to be traversed one by one. Figure 11 illustrates the schematic operation of this algorithm and compares several iterations of the naïve approach with ray tracing. It becomes immediately clear that the number of new cells per iteration remains constant ($3^{(d-1)}$), while the number of cells in the naïve approach grows exponentially per iteration. Additionally, the naïve method works rather in shell layers, including unnecessary cells at the edges of the shell.

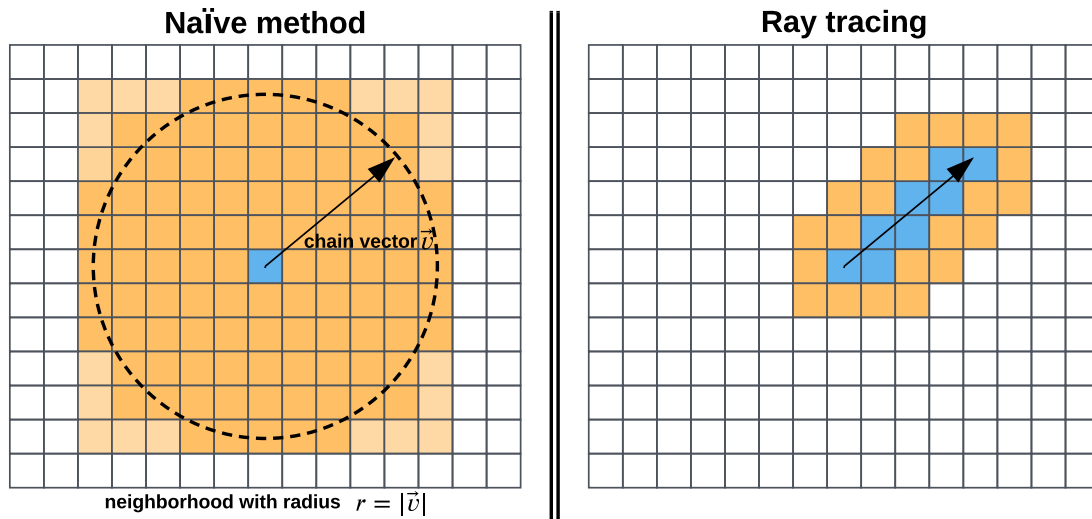


Figure 11: Comparison of the **naïve** method and **ray tracing** in a 2D scenario. In the **naïve** approach, the chain vector is used as a radius to determine the neighborhood. While it can also work iteratively by considering one shell at a time, the number of cells increases exponentially. In this approach, the **blue cell** represents the starting cell. For the **ray tracing** method, the **blue cells** indicate the cells traversed by the ray, while the **orange cells** represent additional cells considered for collision detection.

Complementing Figure 11, the ray tracing algorithm is presented in Algorithm 1, which is essentially a modified version of Bresenham's algorithm for line rasterization [9].

Algorithm 1: Ray tracing algorithm for traversing a grid, motivated by Bresenham's line rasterization algorithm [9].

Input: Starting point **start**, direction vector **dir**, maximum distance $d_{\max}$

Output: Sequence of traversed cells

Initialization

```

dir  $\leftarrow \frac{\mathbf{dir}}{\|\mathbf{dir}\|}$ ;
for  $i \in \{x, y, z\}$  do
     $current\_cell_i \leftarrow \left\lfloor \frac{start_i}{cell\_size_i} \right\rfloor$ ,  $step_i \leftarrow \text{sign}(dir_i)$ ;
end
for  $i \in \{x, y, z\}$  do
     $t_{\max_i} \leftarrow \frac{cell\_size_i \cdot \left\lceil \frac{start_i}{cell\_size_i} \right\rceil - start_i}{dir_i}$  if  $dir_i >$ 
    0 else  $\frac{start_i - cell\_size_i \cdot \left\lfloor \frac{start_i}{cell\_size_i} \right\rfloor}{-dir_i}$ ;
     $t_{\delta_i} \leftarrow \frac{cell\_size_i}{|dir_i|}$ ;
end
 $total\_distance \leftarrow 0$ ;
while  $total\_distance \leq d_{\max}$  and current_cell within the grid do
     $k \leftarrow \arg \min_{i \in \{x, y, z\}} t_{\max_i}$ ;
     $step\_distance \leftarrow t_{\max_k} - total\_distance$ ;
     $total\_distance \leftarrow t_{\max_k}$ ;
     $current\_cell_k \leftarrow current\_cell_k + step_k$ ;
     $t_{\max_k} \leftarrow t_{\max_k} + t_{\delta_k}$ ;
    if current_cellk outside the grid then
        if Periodic boundary condition along axis k then
             $current\_cell_k \leftarrow current\_cell_k \bmod num\_cells_k$ ;
        else
            Break the loop;
        end
    end
end
end

```

3.3.1 Scaling Behavior

In theory, the combination of a grid and ray tracing ensures that the speed of neighbor determination is independent of the system size and the number of particles. This behavior can be verified by determining the average duration per collision for different system sizes at various densities and displacement lengths. Unfortunately, it cannot be entirely avoided that other factors in the implementation also play a role, which is why a constant scaling behavior is not expected.

The benefits of ray tracing can be further explored by examining the number of cells to be checked depending on different systems and the method used. Two brief analyses are performed and combined here.

The first investigation considers the theoretical scaling behavior of the two methods with respect to the number of iterations. From simple geometric considerations, we find that the number of checked cells scales as follows:

$$\begin{aligned} N_1 &= d^3 \\ N_{i,naïve} &= N_1 + ((i+2)^d - (i+1)^d) \\ N_{i,ray} &= N_1 + (d-1)(i-1) \end{aligned} \tag{15}$$

where d denotes the dimension and N_1 represents the initial number of cells considered. For $d = 3$, the initial value is $N_1 = 27$ for both methods. For the second iteration, the increase is given by $N_{2,naïve} - N_1 = 4^3 - 3^3 = 37$ and $N_{2,ray} - N_1 = 9$. Thus, only after the second iteration, ray tracing already considers 28 fewer cells. Furthermore, the number of cells to be checked in the next iteration remains constant for ray tracing, while it grows exponentially for the naïve method.

This behavior can also be verified graphically. Figure 12a illustrates the scaling behavior described in Equations (15) for up to 10 iterations. The exponential growth is evident, as the number of cells checked for ray tracing remains just above 100, while it approaches nearly 1400 for the naïve method.

It remains to relate these numbers to different systems, which results in the second investigation. A significant parameter is the packing fraction of a system. For a system of hard spheres in 3D, the mean free path is given by:

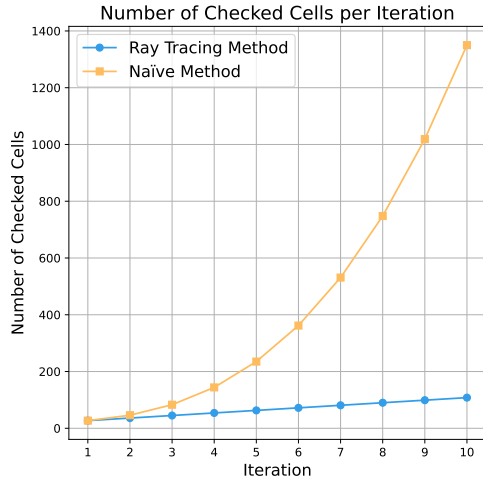
$$\lambda = \frac{1}{\sqrt{2} \cdot \rho \cdot \pi \cdot \sigma^2} \tag{16}$$

where $\rho = \frac{N}{V}$ is the particle number density of the system, and σ is the diameter of the spheres. Using the equation:

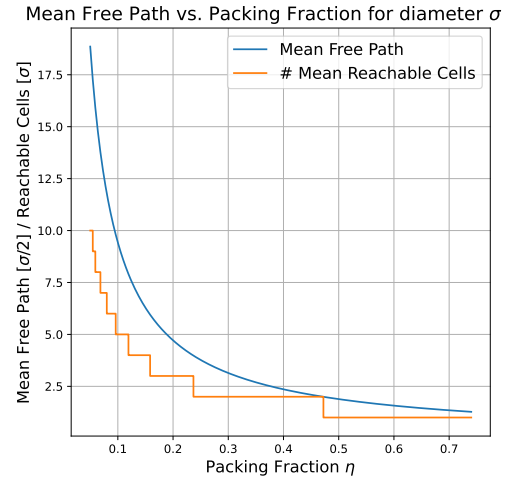
$$\eta = \frac{6\rho}{\pi\sigma^3} \tag{17}$$

the particle density ρ can be translated into the packing fraction η , which is also used for the analysis of phase transitions.

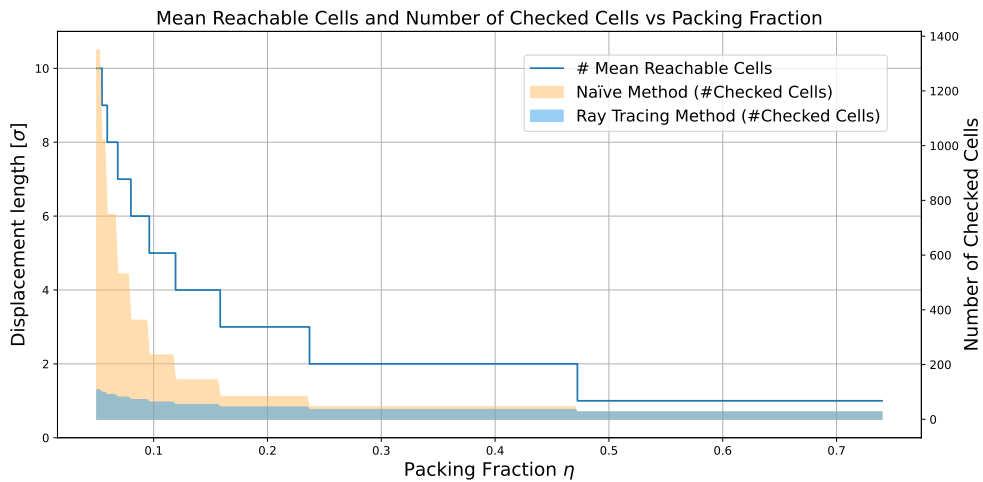
Applying the mean free path λ to the smallest cell side length σ , the number of cells to be checked can be determined from this length and, therefore, from the packing fraction.



(a) Scaling behavior of the number of cells checked by both methods. The exponential growth for the naïve method is evident, while the ray tracing method maintains a constant number of checked cells.



(b) Mean free path and the maximum number of reachable cells as a function of the packing fraction. This plot highlights how the density affects the mean free path and the number of cells to be checked.



(c) Combined scaling behavior and mean free path analyses. It reveals that ray tracing is more efficient at lower densities and higher displacement lengths, while at high densities or short lengths, both methods exhibit similar computational costs.

For the naïve method, this corresponds to a radius. For ray tracing, this number represents more of an upper bound for the maximum number of cells to be checked, since the chain direction can also move diagonally through cells, thus traversing fewer cells for the same length. The lower bound for the number of traversed cells in ray tracing is given by:

$$n_{min} = \frac{n_{max}}{\sqrt{d}} \quad (18)$$

where, for both methods, we use:

$$n_{max} = \left\lceil \frac{\lambda}{\sigma} \right\rceil \quad (19)$$

Figure 12b shows the number of maximum reachable cells and the mean free path as a function of packing fraction. Note that in this implementation, the diameter $\sigma = 2$ was chosen, which causes the scales to shift slightly.

Combining both figures, the packing fraction can be directly translated into the average number of cells to be checked. The combined analysis is presented in Figure 12c. It becomes immediately clear that ray tracing is only efficient at lower densities and higher displacement lengths. For very dense systems or very short displacement lengths, both methods share the same computational cost.

4 Crystallization processes in hard sphere systems [S. Nienhaus]

Hard spheres despite being an easily described system can show complex behavior. Most important are phase transitions which are studied in the following sections. The reasons and theories for these phase transitions are discussed in this section. The focus will be on crystal formation meaning that metrics for detecting crystalline formations need to be introduced in section 5 before actually investigating crystallization processes in a computer simulation in section 6. At this point, it is important to note that from here on, the simulation will no longer be conducted using the author's implementation. xyz-files are generated using polyeventchain [6] to be analyzed again by own implementations of order parameters. The reason for this is that simulating the crystallization processes with the authors implementation takes too much time requiring a more optimized solution. The phase transition from fluid to solid is driven by entropy. Therefore as pointed out by Loewen et al. [10] it is expected that the system would not crystallize but instead always develop into a fluid state because intuitively the fluid state is believed to have higher entropy as it is less ordered. Because a phase transition can be seen this cannot be the case. The solution is noticing there are two types of entropy in this system, a configurational and a free volume entropy. Configurational entropy describes the ability of a fluid system to rearrange spheres in a way that alters their relative positions. The free volume entropy describes the possibility to generate new configurations by simply shifting the existing one a bit. Knowing that it becomes obvious that for the fluid there is a high configurational entropy as there is now order in the system, but a relatively low free volume entropy as the fluid is less dense than the solid. The solid or at first a very dense fluid system has low configurational entropy, because most configurations are forbidden due to overlaps. However, since crystals are typically more compact, crystallization at high densities can increase the free volume, making the crystalline phase the one with higher entropy.

The densities or packing fractions for the fluid to solid transition can be estimated theoretically. To do so a theory for the fluid and a theory for the solid phase can be combined. These theories need to provide an estimate for the free energy of the respective phase. The packing fractions result from the tangent construction at the intersection of these two free energies with additional conditions $p_1 = p_2$ and $\mu_1 = \mu_2$ for the region of coexisting phases where 1 and 2 mark the different theories. In the following the Carnahan-Starling theory describes the fluid phase and the cell theory the solid phase. For the Carnahan-Starling theory one starts with the equation of state to get

$$\frac{pV}{Nk_B T} = -\frac{V}{Nk_B T} \frac{\partial F}{\partial V} = -V f'(\eta) \frac{\partial \eta}{\partial V} \stackrel{\frac{\partial \eta}{\partial V} = \frac{-\eta}{V}}{=} \eta f'(\eta). \quad (20)$$

Integration can now yield the free energy of the fluid phase

$$f_{CS}(\eta) = \text{const}_1 + \int d\eta \frac{1}{\eta} \frac{p_{CS} V}{Nk_B T}(\eta) \quad (21)$$

$$= \text{const}_2 + \ln(\eta) + \int_0^\eta d\tilde{\eta} \frac{1}{\tilde{\eta}} \left(\frac{p_{CS} V}{Nk_B T}(\tilde{\eta}) - 1 \right). \quad (22)$$

The free energy according to the Carnahan-Starling theory still needs to fulfill $f_{\text{CS}}(\eta) \xrightarrow{\eta \rightarrow 0} f_{\text{id}}$. That is why in the last step of Equation 21 the integral is split first separating the $\ln(\eta)$ term present in

$$f_{\text{id}}(\eta) = \frac{F_{\text{id}}}{Nk_{\text{B}}T} = 3 \ln \left(\frac{\Lambda}{\sigma} \right) + \ln \left(\frac{6}{\pi} \right) - 1 + \ln(\eta). \quad (23)$$

The transformation from an indefinite integral to a definite integral simplifies finding const_2 because in the result

$$\begin{aligned} f_{\text{CS}}(\eta) &= f_{\text{id}}(\eta) + \left. \frac{3 - 2\tilde{\eta}}{(1 - \tilde{\eta})^2} \right|_0^\eta \\ &= f_{\text{id}}(\eta) + \frac{4\eta - 3\eta^2}{(1 - \eta)^2} \end{aligned}$$

there is $\lim_{\eta \rightarrow 0} \frac{4\eta - 3\eta^2}{(1 - \eta)^2} = 0$. Therefore the final result for the free energy is

$$f_{\text{CS}} = 3 \ln \left(\frac{\Lambda}{\sigma} \right) + \ln \left(\frac{6}{\pi} \right) - 1 + \ln(\eta) + \frac{4\eta - 3\eta^2}{(1 - \eta)^2} \quad (24)$$

The cell theory starts with the partition function. In General the partition function of a system is given by

$$\begin{aligned} Z_k(T, N, V) &= \frac{1}{h^{3N}N!} \underbrace{\int d^3\vec{q}_1 \dots d^3\vec{q}_N \exp(-\beta V(\{\vec{q}_i\}))}_{\equiv Q(T, N, V)} \int d^3\vec{p}_1 \dots d^3\vec{p}_N \exp \left(-\beta \frac{\sum_{i=1}^N \vec{p}_i^2}{2m} \right) \\ &= \frac{1}{h^{3N}N!} Q(T, V, N) \left(\int d^3\vec{p} e^{-\frac{\beta \vec{p}^2}{2m}} \right)^N \end{aligned}$$

where $Q(T, V, N)$ is the configuration integral [11]. The cell model now assumes that in the solid phase every sphere is fixed at its lattice position and can only move within the free volume around the lattice position which is usually described using a Wigner-Seitz cell. In the cell model the Wigner-Seitz cell is replaced by a sphere that models the free volume. These assumptions together with

$$\frac{1}{h^3} \int d^3\vec{p} e^{-\frac{\beta \vec{p}^2}{2m}} = \frac{1}{\Lambda^3} \quad (25)$$

can be used to calculate the partition function

$$Z_{\text{cell}} = \frac{v_f^N}{\Lambda^{3N}}.$$

$\Lambda = \sqrt{\frac{\beta h^2}{2\pi m}}$ is the De Broglie wavelength. It has been used that the configuration integral is $Q(T, V, N) = N!v_f^N$. The factor $N!$ results from the fact that every sphere could be

located in every lattice position. v_f as explained before is approximated by a sphere whose radius is determined by the distance of spheres in the lattice r and their diameter σ . The resulting

$$v_f = \frac{4\pi}{3}(r - \sigma)^3$$

can be related to the packing fraction $\eta_{\text{cp}} = \frac{\pi}{3\sqrt{2}}$ for dense packing where $r = \sigma$. Taking the general formula $\eta = \frac{\pi}{6}\rho\sigma^3$ and the one for dense packing $\eta_{\text{cp}} = \frac{\pi}{6}\rho r^3$ the relation

$$\frac{r}{\sigma} = \left(\frac{\eta_{\text{cp}}}{\eta} \right)^{1/3}$$

can be derived and inserted to get

$$v_f = \frac{4\pi}{3}\sigma^3 \left(\frac{r}{\sigma} - 1 \right)^3 = \frac{4\pi}{3}\sigma^3 \left(\left(\frac{\eta_{\text{cp}}}{\eta} \right)^{1/3} - 1 \right)^3.$$

Now using the known relation between free energy and partition function

$$F_{\text{cell}} = -k_{\text{B}}T \ln(Z_{\text{cell}})$$

the free energy in the cell model is calculated as

$$f_{\text{cell}}(\eta) = 3 \ln \left(\frac{\Lambda}{\sigma} \right) - \ln \left(\frac{4\pi}{3} \right) - 3 \ln \left(\left(\frac{\eta_{\text{cp}}}{\eta} \right)^{1/3} - 1 \right). \quad (26)$$

Calculating the intersection of the free energies Equation 24 and Equation 26 yields the packing fraction $\eta = 0.544$. The tangent construction at this intersection point is shown in Figure 13.

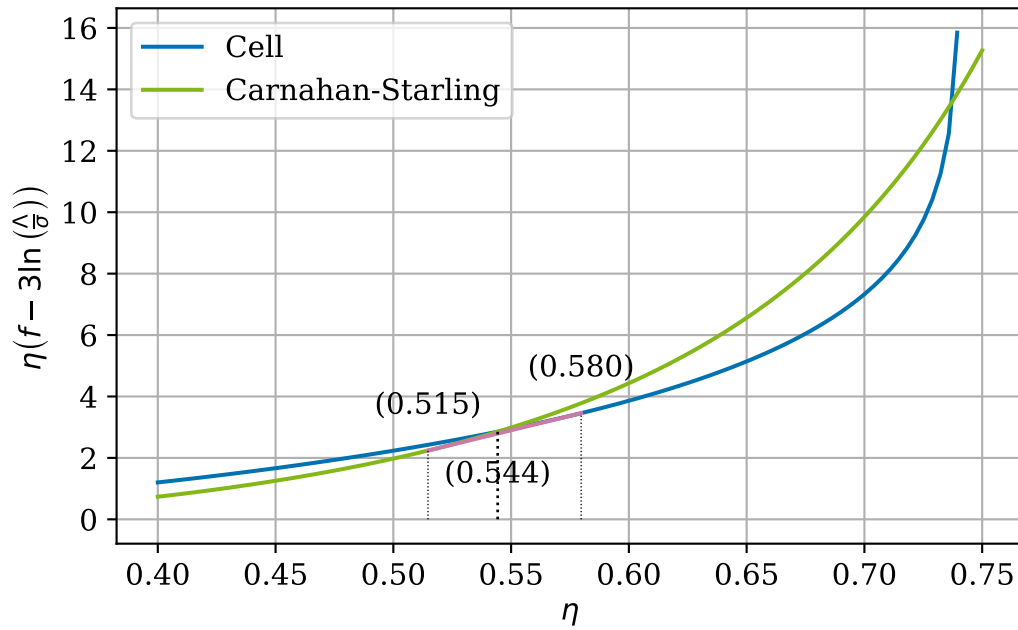


Figure 13: Shown are the free energies according to the Carnahan-Starling- and cell theory. With a tangent at the intersection the coexistence region has been determined and marked with vertical lines.

To get to η_f and η_s the conditions

1. $p_{cs}(\eta_f) = p_{cell}(\eta_s)$
2. $\mu_{cs}(\eta_f) = \mu_{cell}(\eta_s)$

are used. 1. can be transformed with Equation 20. Condition 2. can be transformed with

$$\begin{aligned}
 a) \quad & F = Nk_B T f(\eta) \\
 b) \quad & \mu = \frac{\partial F}{\partial N} = \frac{F}{N} + Nk_B T \frac{v_s}{V} f'(\eta) \\
 \Rightarrow \quad & \frac{\mu}{k_B T} = f(\eta) + \eta f'(\eta).
 \end{aligned}$$

The resulting system of equations

$$1. \quad \eta_f^2 f'_{CS}(\eta_f) = \eta_s^2 f'_{cell}(\eta_s) \quad (27)$$

$$2. \quad f_{cs}(\eta_f) + \eta_f f'_{cs}(\eta_f) = f_{cell}(\eta_s) + \eta_s f'_{cell}(\eta_s) \quad (28)$$

needs to be solved for η_f and η_s . The results are $\eta_f \simeq 0.515$ and $\eta_s \simeq 0.580$. [1]

The previous calculations have shown one of the simpler ways to obtain an estimate for the densities of the coexisting phases. Therefore the results diverge from the actual

values of $\eta_f = 0.494$ and $\eta_s = 0.545$ [10]. These results can be obtained with more complex theories like density functional theory too, but to this point simulations are still considered to give the most exact description of the hard sphere system. These simulations have also uncovered the existence of a glass phase as explained in Figure 14.

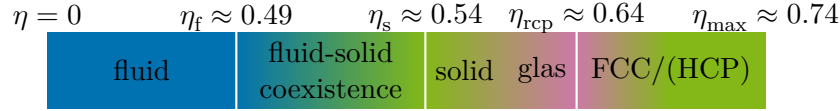


Figure 14: Sketch of the phase transitions appearing in hard sphere systems. (adapted from [12])

In the glass phase due to the density of the system, dynamics slow down so much that no crystallization occurs for a very long time and the system stays in an unordered state. In these simulations it can also be observed that the FCC structures are the most dominant and HCP is less common. Going back to the explanation from the beginning where the free volume entropy has been used to explain the crystallization one would expect both to be equally common as they have the same maximum packing fraction. However, simulations have shown that the FCC structure has slightly higher entropy, resulting in an entropy difference of $\Delta S = 0.001164(8)Nk_B$ [13].

5 Order parameters [S. Nienhaus]

This work will deal with the q_l/Q_l order parameters with a focus on $l = 6$. Using this order parameter an additional q_6q_6 parameter can be constructed which will be applied to identify the solid phase. The minkowski tensor can be applied after the q_6q_6 to determine which type of crystal is present.

5.1 q_l [S. Nienhaus]

The local q_l and global Q_l order parameters are given by

$$q_l = \frac{1}{N} \sum_{i=1}^N \sqrt{\frac{4\pi}{2l+1} \sum_{m=-l}^{m=l} \left| \frac{1}{n(i)} \sum_{j=1}^{n(i)} Y_{l,m}(\mathbf{r}_{ij}) \right|^2} \quad (29)$$

$$Q_l = \sqrt{\frac{4\pi}{2l+1} \sum_{m=-l}^{m=l} \left| \frac{1}{Nn(i)} \sum_{i=1}^N \sum_{j=1}^{n(i)} Y_{l,m}(\mathbf{r}_{ij}) \right|^2} \quad (30)$$

where $n(i)$ is the number of nearest neighbors (nn) of particle i and $\mathbf{r}_{ij}$ is the vector pointing from particle i to particle j . $Y_{l,m}$ are the spherical harmonics of order l . N is the total number of particles [14].

The basic idea behind simple order parameters is to weight the positions of the nearest Neighbors of each particle in such a way that the phases of interest get high values. In the case of Q_l , the spherical harmonics of order l determine the weights. In most cases $l = 6$ is chosen. Why are weights with $l = 6$ a good choice depending on the task ? As shown in Figure 15 Q_6 returns high values for the most important (for the hard sphere crystallization) crystal structures FCC and HCP compared to other orders. Additionally q_6 is sensitive to icosahedral crystal structures.

As explained in [16] q_6 still has some shortcomings. It depends on how the nearest neighbors are chosen and therefore is not uniquely defined. In this work the number of nn is fixed and chosen to be 12. Other works like the original one by Steinhardt et al. [15] have chosen nn based on a cutoff. Therefore values of different papers cannot be compared directly.

It can be argued that q_6 on its own is not quite meaningful. For finding the solid phase and especially for finding specific crystalline structures Figure 16 gives a hint to an issue concerning background contributions. There are clearly visible peaks for the most interesting crystal types but these sit on top of a fluid background. Because of the background a q_6 value being within one of the peaks does not necessarily mean that the particle is part of that crystal structure. For particles having values within the HCP peak region it is even more likely they are still being part of the fluid phase.

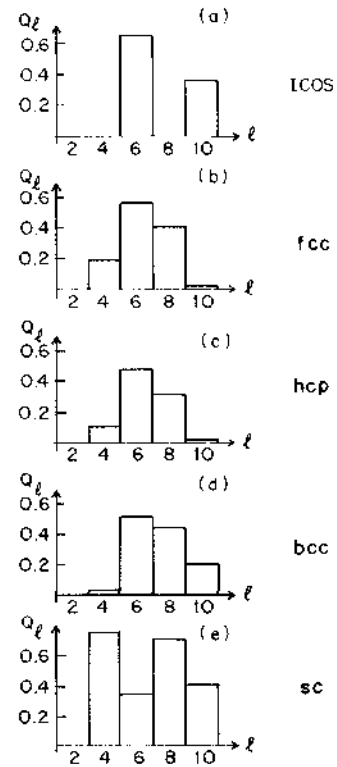


Figure 15: Results of different orders of Q_l for different crystals. Note that the nearest neighbors here are chosen by defining a cutoff radius. [15]

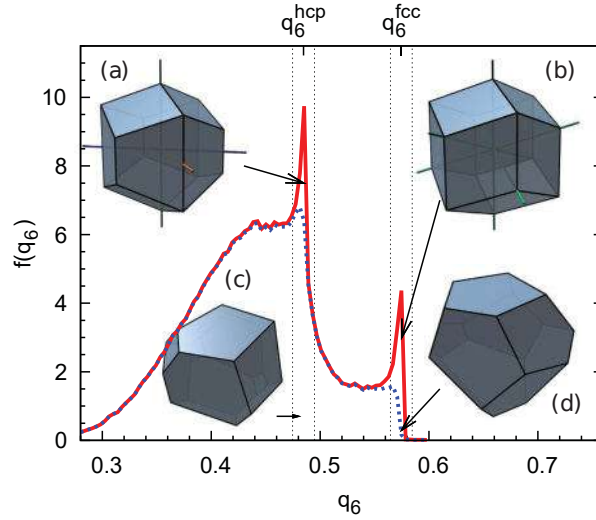


Figure 16: The red line gives the abundances of all cells with the given value of q_6 . The blue line marks the particles that are neither FCC nor HCP based on the $\Delta_{\text{FCC/HCP}}$ as explained in subsection 5.3 [17].

5.2 q_6q_6 [S. Nienhaus]

A structure metric is needed, that is better at identifying the solid phase. To do so ten Wolde et al. [18] introduced the q_6q_6 order parameter. It is based on the q_6 order parameter and therefore is still dependent on the definition of nn. However, due to including information about the q_6 values of nn it performs better for identifying the solid phase. q_lq_l is defined using

$$\overline{Y_{lm}}(i) = \frac{1}{N_i} \sum_j^{N_i} Y_{lm}(\theta(r_j), \phi(r_j)) \quad q_{lm}(i) = \frac{\overline{Y_{lm}}(i)}{\sqrt{\sum_{m=-l}^l |\overline{Y_{lm}}(i)|^2}}$$

as

$$q_lq_l(i) = \sum_j^{N_i} \sum_{m=-l}^l q_{lm}(i)q_{lm}^*(j) \quad [12].$$

Particles that exceed a certain threshold are considered to be part of the solid phase. The q_6q_6 still should not be used as a structure metric for direct identification of FCC or HCP as can be seen in Figure 21, because distributions will show peaks indicating FCC and HCP contributions, but these peaks do not occur at distinct values that can be directly linked to a specific crystal type. Nevertheless for identifying the solid this plot shows a better separation between the fluid initial state and the much more ordered intermediate state in contrast to the results for the q_6 parameter where the background problem can be seen in Figure 18.

5.3 Minkowski tensors [S. Nienhaus]

Minkowski tensors can solve the dependence on the nn definition too. The basic idea is to not only take bonds to nn but also weight the contribution of these bonds by their associated Voronoi cell face area. These ideas result in

$$(W_1^{0,4})_{ijkl} = \sum_{i=1}^{N_i} a(i) n_i n_j n_k n_l \quad (31)$$

which is the rank four minkowski tensor. N_i is the number of faces of the Voronoi cell of that particle, $a(i)$ is the area of the i -th face and n_j is a component of the normal vector of that face.

The rank-four Minkowski tensor belongs to a broader class of tensors that originate from minkowski functionals. These functionals, defined as

$$W_0(K) := \int_K d^2r \quad \text{and} \quad W_\nu(K) := \frac{1}{2} \int_{\partial K} G_\nu dr$$

are used as shape metrics but in contrast to minkowski tensors they are not able to measure anisotropy and orientation. K is a convex body with a smooth boundary contour ∂K . G_ν is a symmetric combination of the principal curvatures κ_i of the boundary ∂K . Therefore for the minkowski tensors the normals $\vec{n}$ of the boundary contour ∂K are included, making the definition

$$W_0^{a,0}(K) := \int_K \vec{r}^{\odot a} d^2r \quad (\nu = b = 0) \quad (32)$$

$$W_\nu^{a,b}(K) := \frac{1}{2} \int_{\partial K} \vec{r}^{\odot a} \odot \vec{n}^{\odot b} G_\nu dr \quad (\nu, b \neq 0) \quad (33)$$

sensitive to anisotropies and the orientation. The definition above includes tensor products using the notation

$$\vec{r}^{\odot a} \odot \vec{n}^{\odot b} = \underbrace{\vec{r} \odot \dots \odot \vec{r}}_{a \text{ times}} \odot \underbrace{\vec{n} \odot \dots \odot \vec{n}}_{b \text{ times}}.$$

All minkowski tensors have three properties in common being motion covariance, additivity and being conditionally continuous. Conditionally continuous means that small changes of the body K will not result in large changes of the minkowski tensor. In Equation 31 for example this is the case due to the weighting with the face areas. When the neighborhood changes slightly leading to a new face normal of the voronoi cell to appear this will not have a large impact as this new face normal is weighted by a small face area.

Additivity is defined as

$$W_\nu^{a,b}(K_1 \cup K_2) + W_\nu^{a,b}(K_1 \cap K_2) = W_\nu^{a,b}(K_1) + W_\nu^{a,b}(K_2).$$

This implies the possibility to analyze the structure of larger regions or for a collection of a couple of spheres by simply adding the minkowski tensors for smaller subsets of that region or for the contained spheres.

While the minkowski functionals are motion invariant, minkowski tensors are motion covariant with transformation properties

$$W_\nu^{a,b}(K \uplus \mathbf{t}) = \sum_{c=0}^a \mathbf{t}^{\odot c} \odot W_\nu^{a-c,b}(K) \quad (34)$$

$$W_\nu^{a,b}(\hat{O}K) = \hat{O}W_\nu^{a,b}(K) \quad (35)$$

for translations $\mathbf{t}$ and rotations $\hat{O}$. This means for the rank four minkowski tensor from Equation 31 that it is translation invariant which is a required property for the following analysis. However, it still is not rotation invariant.[17]

To get from the general definition Equation 32 for a general convex body K to the definition in Equation 31 with the voronoi cell instead of K , a parallel body construction needs to be performed as shown in [17, pp. 56–58].

In the following only the rank four minkowski tensor will be used to identify the crystal types. The tensor $W_1^{0.4}$ is normalized by the sum of its diagonal elements, which equals the total area of all Voronoi cell areas [17].

$$S_{ijkl} = (W_1^{0.4})_{ijkl} / (\sum_{o,p} (W_1^{0.4})_{oopp})$$

The tensor S_{ijkl} has three symmetries

$$S_{ijkl} = S_{jikl} \quad S_{ijkl} = S_{ijlk} \quad S_{ijkl} = S_{klij}$$

which allow to reduce the 81 elements large tensor to a 6x6 matrix also known as the construction of a voigt matrix. As mentioned before the minkowski tensor still needs to be made rotationally invariant. To make this matrix rotationally invariant some factors need to be added as well to get to

$$\mathbf{M}(S) \equiv \begin{bmatrix} S_{xxxx} & S_{xxyy} & S_{xxzz} & \sqrt{2}S_{xxyz} & \sqrt{2}S_{xxzx} & \sqrt{2}S_{xxxy} \\ S_{xxyy} & S_{yyyy} & S_{yyzz} & \sqrt{2}S_{yyyz} & \sqrt{2}S_{yyxz} & \sqrt{2}S_{yyxy} \\ S_{xxzz} & S_{yyzz} & S_{zzzz} & \sqrt{2}S_{zzyz} & \sqrt{2}S_{zzxz} & \sqrt{2}S_{zzxy} \\ \sqrt{2}S_{xxyz} & \sqrt{2}S_{yyyz} & \sqrt{2}S_{zzyz} & 2S_{yzyz} & 2S_{yzxz} & 2S_{yzxy} \\ \sqrt{2}S_{xxzx} & \sqrt{2}S_{yyxz} & \sqrt{2}S_{zzxz} & 2S_{yzxz} & 2S_{xzxz} & 2S_{xzxxy} \\ \sqrt{2}S_{xxxy} & \sqrt{2}S_{yyxy} & \sqrt{2}S_{zzxy} & 2S_{yzxy} & 2S_{xyxz} & 2S_{xyxy} \end{bmatrix}$$

The eigenvalues of $\mathbf{M}$ are calculated. These eigenvalues can already be used to identify crystal structures. To do so the eigenvalues of these crystals as given in Table 3 need to be known.

Table 3: Eigenvalues of the FCC and HCP crystal.

	FCC	HCP
M_1	$\frac{1}{3}$	$\frac{1}{3}$
M_2	$\frac{1}{6}$	$\frac{1}{6}$
M_3	$\frac{1}{6}$	$\frac{2}{15}$
M_4	$\frac{1}{6}$	$\frac{2}{15}$
M_5	$\frac{1}{12}$	$\frac{1}{9}$
M_6	$\frac{1}{12}$	$\frac{1}{9}$

Using the eigenvalues directly proofs to be inconvenient in practice. That is why similar to the mean squared error the deviation of eigenvalues from the crystal of interest is calculated using

$$\Delta_{\text{FCC/HCP}} = \sqrt{\sum_{i=1}^6 [M_i(V) - M_i(V_{\text{FCC/HCP}})]^2}.$$

$M_i(V)$ are the calculated eigenvalues and $M_i(V_{\text{FCC/HCP}})$ are the literature eigenvalues as given in Table 3. $\Delta_{\text{FCC/HCP}}$ can now be used to identify FCC and HCP when given a threshold. This threshold needs to take the slight movement of spheres during the simulation into account. These movements lead to an distribution of $\Delta_{\text{FCC/HCP}}$ even for nearly perfect crystals. This distribution can be fitted with a Gaussian distribution to be able to determine the threshold methodically based on the standard deviation and mean of that distribution. Due to time constraints this will not be done here. In [12] it is shown how to do this properly.

6 Application of order parameters in the study of phase transitions [S. Nienhaus]

First all order parameters are explored individually. At the end of this section all of them are combined to first distinguish between solid and fluid with q_6q_6 and then to assign crystal types to the found solid phase particles using the minkowski structure metric. Most parts of this analysis will work with the same set of simulation data whose parameters are given in Table 4. Before looking into the combined analysis which will include exploring the development of the system over many configurations only three configurations are analyzed for the individual

Table 4: Simulation parameters for exploring the order parameters individually.

L	20
σ	1
l	20
η	0.6

order parameters. These will be the first configuration after using the rattle algorithm to initialize the system, the last configuration after $6 \cdot 10^7$ sweeps and an intermediate configuration after 10^7 sweeps. We can use the common neighbor analysis in ovito [19] to visualize all three of these configurations to compare the results of ovito with the results obtained by the order parameters.

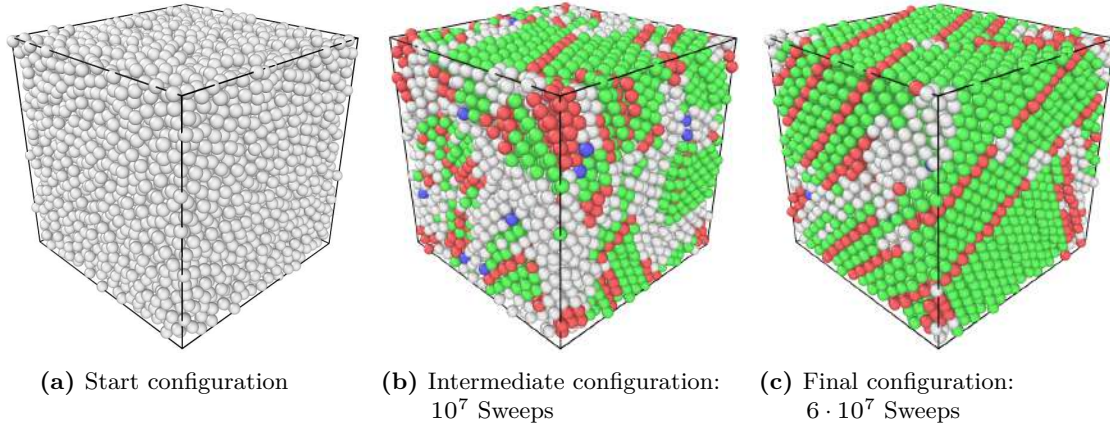


Figure 17: States of the simulation for $\eta = 0.6$ at different times.

6.1 q_l [S. Nienhaus]

Figure 18 shows the distribution of q_6 values for the first and the last simulation state. In the first state the system is in a fluid phase. Therefore values for q_6 are distributed randomly around ≈ 0.4 . In the final state the system has crystallized into FCC and HCP. The distribution of q_6 now shows two peaks corresponding to the two crystal formations. Vertical lines mark the values for a perfect FCC and HCP crystal respectively. Vertical lines and peaks do not align due to the small size of the system and the resulting large interfaces between FCC and HCP crystals. That proves once more that one q_6 value on its own contains next to no information.

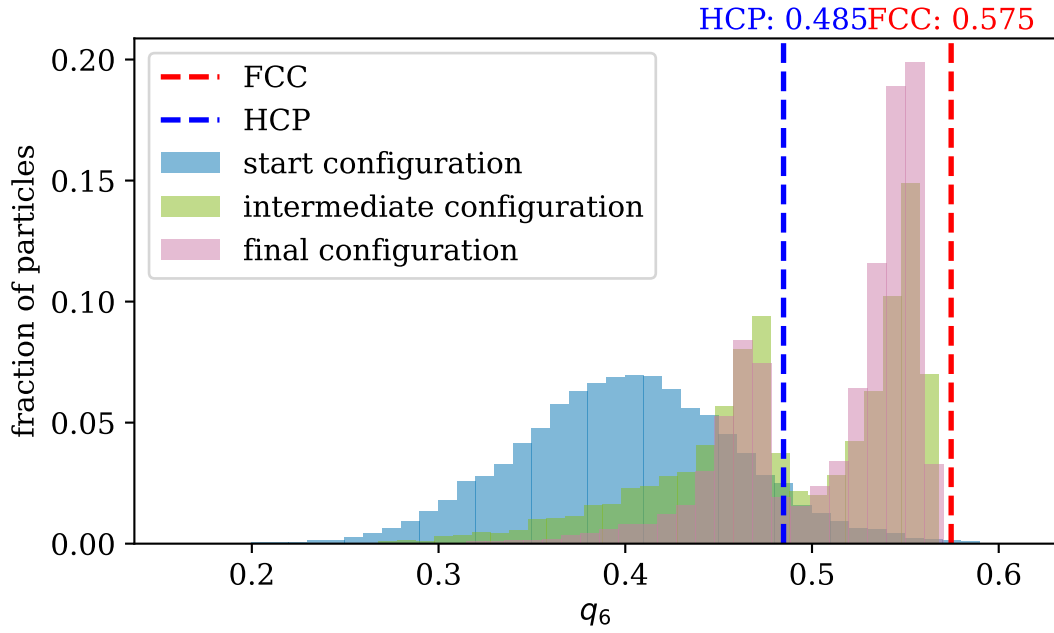
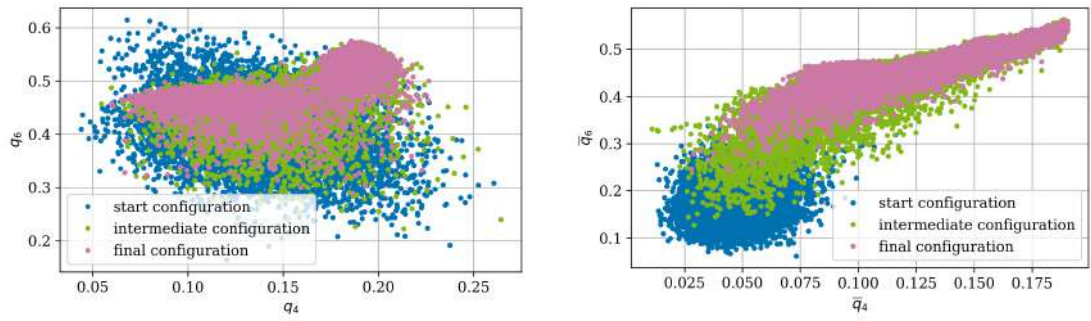


Figure 18: Distribution of q_6 values for three simulation configurations.

That is why q_4 is often used in combination with q_6 as shown in Figure 19a. When comparing Figure 19a with Figure 15 it becomes clear that q_4 can help in separating clusters of different crystals. The issue with having the simple q_6 against q_4 diagram is that the fluid cannot be separated properly from the crystalline particles. Modifying q_6 and q_4 by averaging over the the neighboring particles solves this issue as visible in Figure 19b. The new definition for $\bar{q}_l$ is

$$\bar{q}_l = \sqrt{\frac{4\pi}{2l+1} \sum_{m=-l}^{m=l} \left| \frac{1}{N_{\text{nb}} n(i)} \sum_{i=1}^{N_{\text{nb}}} \sum_{j=1}^{n(i)} Y_{l,m}(\mathbf{r}_{ij}) \right|^2} \quad [20]$$

When compared with Equation 29 this is just the global Q_6 parameter, but the averaging sum runs only over the nearest neighbors instead of over the whole system.



(a) Different configurations displayed in the q_6 - q_4 plane. (b) Different configurations displayed in the $\bar{q}_6$ - $\bar{q}_4$ plane.

Figure 19: On the left the different phases are separated badly by the simple q_6 - q_4 diagram. On the right the separation is much better using the averaged order parameters.

The additional separation of different crystals is not clearly visible in Figure 19b. This might be the case due to small a system size. The literature in contrast can clearly show the separation of all phases using the averaged q_l in Figure 20.

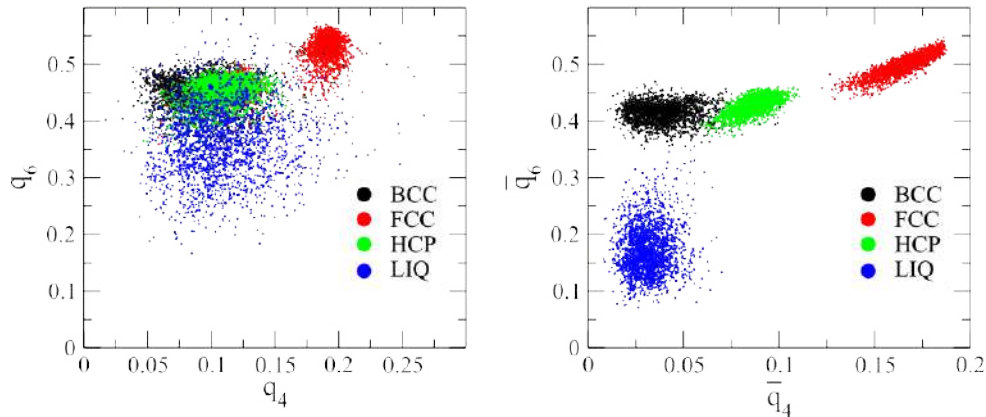


Figure 20: Separation of different crystal types using $\bar{q}_6$ and $\bar{q}_4$ [20].

6.2 q_6q_6 [S. Nienhaus]

Figure 21 shows the distribution of q_6q_6 values for the first, the last and an intermediate simulation state.

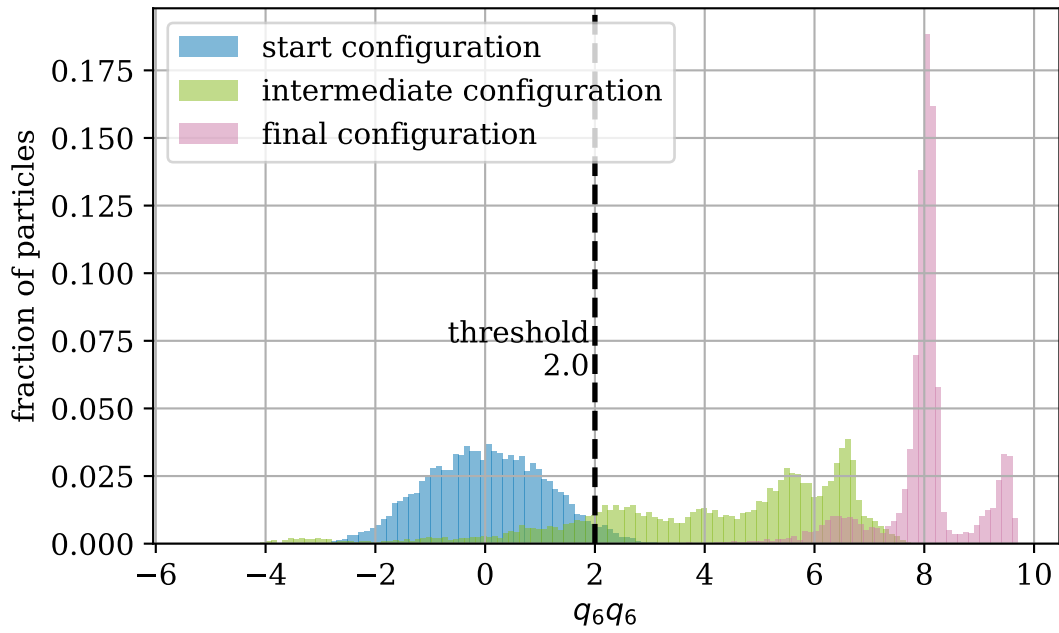
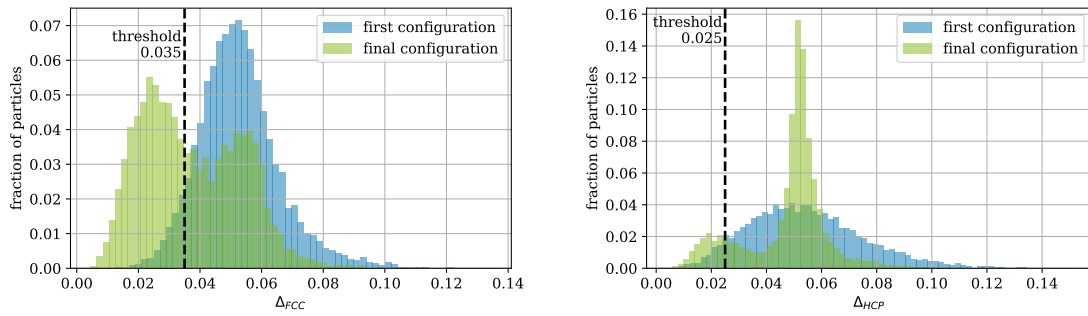


Figure 21: Distribution of q_6q_6 values for three different configurations.

To use q_6q_6 a threshold needs to be determined to distinguish between solid and fluid. This threshold should make sure to exclude most of the first configuration from the solid phase but not to exclude too many crystal particles with more fluid surroundings from the solid phase. Therefore the threshold is set to 2.

6.3 Minkowski tensors [S. Nienhaus]

Figure 22 displays the distributions of $\Delta_{\text{FCC/HCP}}$ for the first and last configuration. Just like for q_6q_6 these distributions can be used to find a threshold for the identification of the crystal type. It can be seen that again two peaks are forming with the one the left belonging to the crystal whose Δ is observed in that diagram. The threshold can be chosen by trying to optimally separate the FCC and HCP peak from the rest of the distribution. Here the threshold could be chosen to be 0.035 and 0.025. Yet at least for HCP this is in fact a bad choice since the fluid phase and the HCP peak overlap too much. In fact, in practice it is better not to choose any threshold at all. In subsection 6.4 q_6q_6 will be used to first discriminate between solid and fluid. After identifying the solid particles their crystal type is determined by the rule $\min(\Delta_{\text{FCC}}, \Delta_{\text{HCP}})$ making a threshold unnecessary.



(a) Distribution of Δ_{FCC} for the first and last configuration. (b) Distribution of Δ_{HCP} for the first and last configuration.

Figure 22: $\Delta_{\text{FCC/HCP}}$ distributions for the first and last configuration. The intermediate state is ignored for the sake of clarity.

6.4 Combination of order parameters [S. Nienhaus]

Now all order parameters are combined for investigating the development of the complete system over many configurations in time. Figure 23 shows the development of the order parameters over time for the same system as has been explored before so for $\eta = 0.6$.

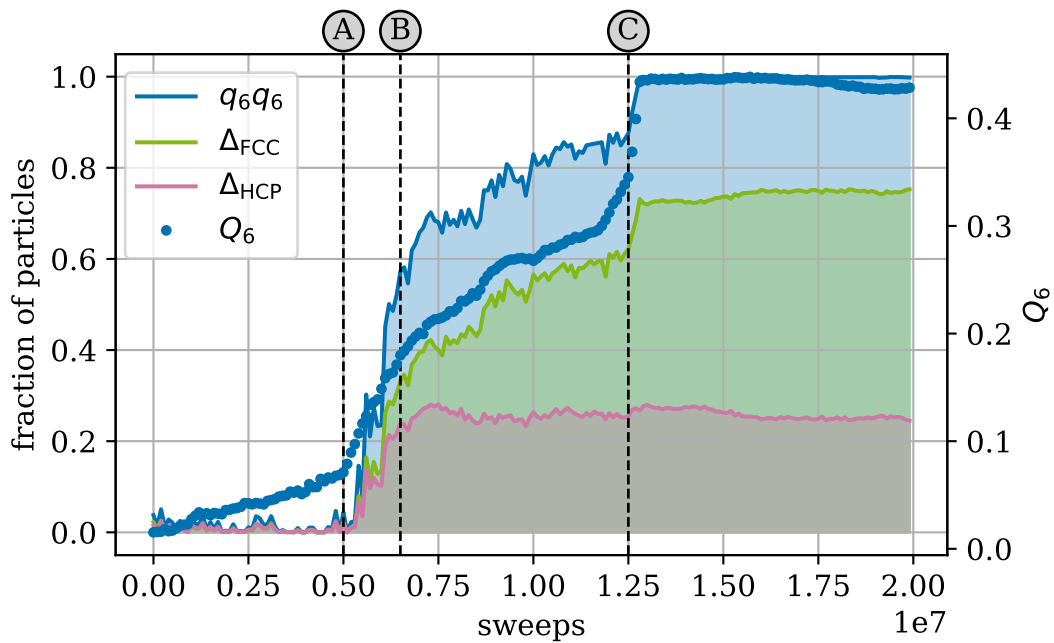


Figure 23: Complete structure analysis for $\eta = 0.6$ using q_6q_6 , Q_6 and the minkowski structure metric.

The results from Figure 23 are compared with snapshots from the common neighbor

analysis in ovito [19]. These snapshots can be found in the appendix. In the first part of the simulation from the start to **A** there is almost no crystallization activity according to q_6q_6 . Nonetheless in the ovito snapshot 49b the development of crystallization cores is shown and $\approx 10\%$ of particles are marked as solid. It is assumed that the reason for that discrepancy is the nearest neighbor definition for the q_6q_6 . One possible explanation is that having 12 neighbors causes smaller crystalline cores to go undetected as they are averaged out among the 12 partially fluid neighbors. This assumption is supported by the analysis in Figure 24 where at first one large crystal develops instead of many small ones (Figure 50b, Figure 50c). This one is detected by q_6q_6 but still the exact fraction of solid particles does not match the data from ovito. In conclusion, the q_6q_6 parameter is less sensitive to small domains compared to ovito's common neighbor analysis, making it more suitable for detecting larger crystal domains. However, it is less effective for determining the precise degree of order in the system, as smaller, partly ordered regions tend to be overlooked.

In contrast, the Q_6 parameter is designed to capture the overall order of the entire system, making a combination of both order parameters valuable for a more comprehensive analysis in the following. In Figure 23 according to Q_6 there indeed is some ordering activity from start to **A** due to some tiny crystallization cores forming. These tiny cores do not form larger crystal domains yet resulting in the vanishing values for q_6q_6 . From **A** to **B** some of these tiny crystallization cores start to grow to form larger crystal domains as can be seen in Figure 50b and Figure 50d now resulting in an increase of Q_6 as well as q_6q_6 . The sudden crystal growth stops between **B** and **C** as only FCC keeps growing and HCP stays nearly constant due to its entropic disadvantage against FCC. Right before **C** crystals coexist with larger fluid phase parts (Figure 49e). At **C** a lot of these fluid parts rapidly turn into FCC leading to the last steep rise in the FCC fraction. From now on the general crystal fraction will not change noticeably anymore and the system has fully solidified.

The next system is simulated for $\eta = 0.54$ to explore the system in the phase coexistence region. Exploring crystallization processes right in the middle of the phase coexistence region or just for slightly lower packing fractions like $\eta = 0.53$ is impractical as the system already takes too long to crystallize.

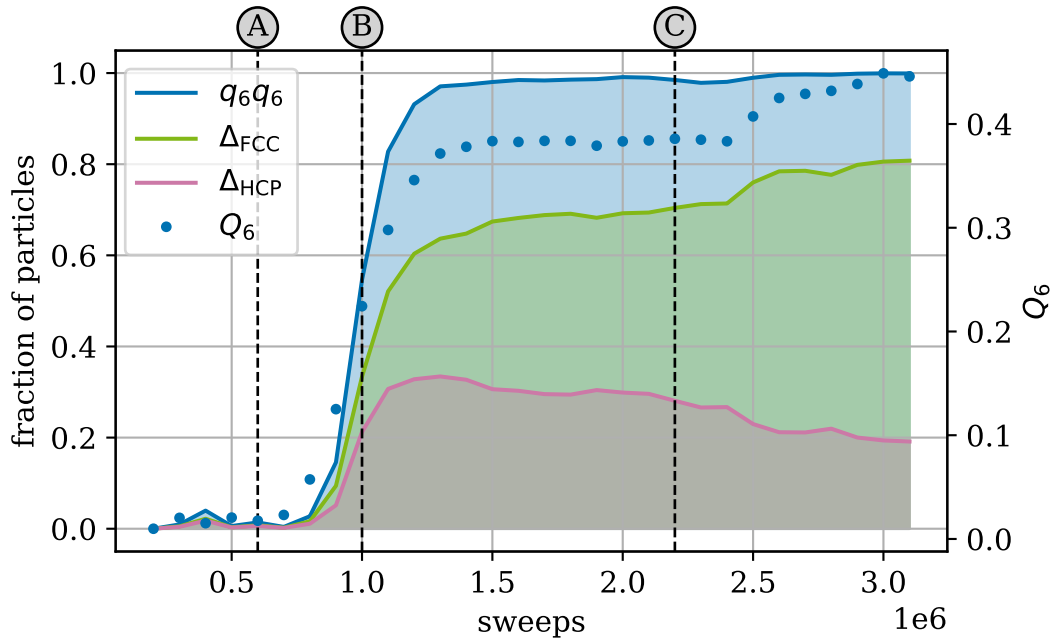


Figure 24: Complete structure analysis for $\eta = 0.54$ using q_6q_6 , Q_6 and the minkowski structure metric.

The simulation in the coexistence region behaves differently. Here crystallization cores do directly grow as soon as they are created. This starts to happen with the first core in Figure 50b and Figure 50c. In the previous simulation it took some time before these cores were large enough to actually grow and to have an impact on q_6q_6 . The growth continues until (B) (Figure 50d) where the competition between FCC and HCP starts to control the system's behavior yet again and the total fraction of particles in the solid phase stays approximately constant. At (C) (Figure 50e) the FCC shows a sudden growth at the expense of the HCP phase. After that the system does not change noticeably anymore.

6.5 Summary of the work on order parameters [S. Nienhaus]

The commonly used q_l and Q_l order parameters have been introduced, along with the possibilities for their individual and combined applications, particularly with q_6 , q_4 , and their averaged versions. The limitations of using these parameters individually were explained. To address some of these shortcomings, the q_6q_6 order parameter was introduced, which is better suited for detecting the solid phase. Additionally, the Minkowski structure metric was implemented to identify the crystal type of the detected solid particles. Finally, it was demonstrated how to integrate all the described parameters for a comprehensive analysis of the crystallization process.

6.6 Melting in 3D [S. Nienhaus]

While only the crystallization process has been studied in the previous chapter the reverse process of melting can appear as well. To explore melting a perfect FCC crystal block is constructed as initial configuration and put into a $L = 20$ simulation volume to reach a packing fraction of $\eta = 0.49$. The chain length of the following simulation is $l = 20$. The authors implementation is used here. In this implementation no definition of a sweep exists. Therefore the number of performed chains is used as a measure for the progress of the simulation. The melting process is monitored by the global Q_6 order parameter in Figure 25.

Melting in three dimensions is a first order phase transition. Meaning that there is a continuous decrease in the order parameter Q_6 until the system has gone from the FCC cube (Figure 51a) to a totally unordered fluid state (Figure 51c) without any intermediate phases.

In the following sections it will be shown that this simple description does not apply to the two-dimensional case. Here a new so called „hexatic“ phase can be observed as an intermediate state in the melting process. Furthermore the solid phase does exist but it is no crystal with positional and orientational order like it is for the three dimensional case.[22]

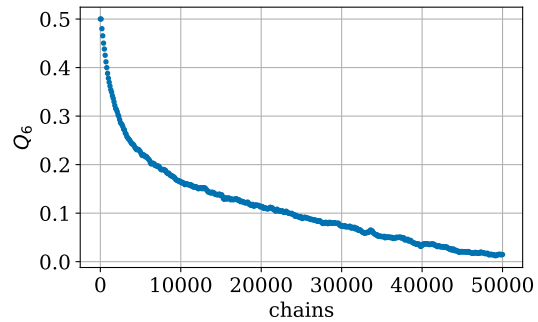


Figure 25: Observation of melting in a 3D System using the global Q_6 parameter.

7 Kosterlitz-Thouless transition in hard disk system [L. Annuss]

7.1 Different phases in two dimensions [L. Annuss]

The two-dimensional hard sphere system was one of the first cases in which the Monte-Carlo method was used to calculate the equation of state for a thermodynamic system with the help of early computers. In this study from the year 1953 Metropolis *et al.* found no evidence of a phase transition in the hard sphere system [23]. A later study in 1962 found evidence for a transition between a fluid and a crystalline phase [24]. This result contradicted the belief that a two-dimensional system is not able to form a crystal. One can show through theoretical considerations that in a two-dimensional system without long-range attractive interaction, it is not possible to create a stable crystal. This is a consequence of thermal fluctuations, which reduce the positional order of the system. An atom or sphere in a three-dimensional system has normally more neighbors than a disk in two dimensions, which makes it more likely that the thermal fluctuations are averaged out. From this, follows that the positional order does not decrease over distance. This stabilizing effect is not present in a two-dimensional system [25]. However, this does not prevent the hard sphere system from forming a solid. An overview of the positional and

orientational order of different phases is given in Table 5.

Table 5: Classification of different phases by positional and orientational order.

Phase	positional order	orientational order
crystalline	long-range	long-range
solid	algebraic	long-range
hexatic	short-range	algebraic
liquid	short-range	short-range

For a long time, it was not clear which process was responsible for melting. One possibility would be that imperfections spread from the grain boundaries and thus start the melting process. According to the Kosterlitz-Thouless theory a two-dimensional solid starts melting by the formation of bound dislocations. These dislocations consist of four disks. The bonds of the dislocated disk deviate from the normal bonding behavior of the solid. In a solid each disk has six neighbors, which form a hexagon around the disk considered. A configuration without lattice defects can be seen in Figure 26a. A bounded dislocation can be seen in Figure 26b. Here two disks, which normally lie diagonally to each other form an additional bond. Now, two of the four disks have seven bonds. This new bond breaks the bond between the other two disks. They now have five neighbors. These kinds of dislocations reduce the positional order to only short-range correlation and the orientational order to an algebraic correlation. This gives rise to a hexatic phase.

The order of the hexatic phase further decreases in the melting process, when the dislocations unbound, as can be seen in Figure 26c. For their work [26] David J. Thouless and J. Michael Kosterlitz received the physic Nobel Prize in 2016.

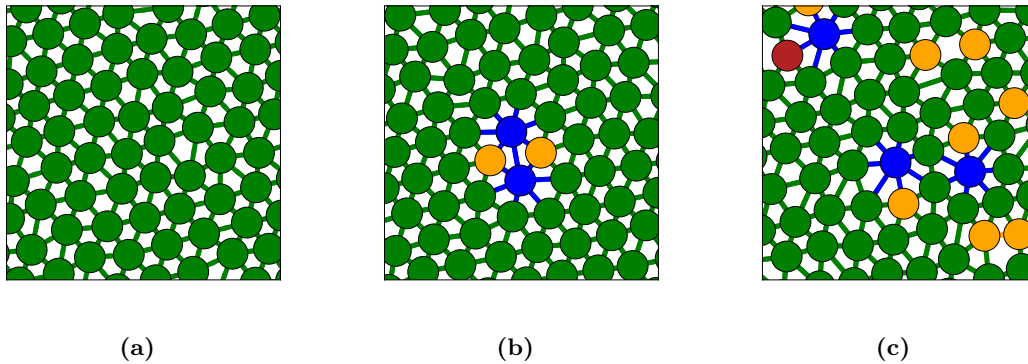


Figure 26: Three cut-outs from a configuration at $\eta = 0.7185$, which shows the steps of a Kosterlitz-Thouless phase transition. The disks are colored by the number of their bonds. A disk with seven neighbors is blue, a disk with six is green, a disk with five is orange, and a disk with four neighbors is colored red. A connection always has the color of the neighboring sphere with the most connections.

To get an insight into the transition described by Kosterlitz-Thouless two-dimensional hard disk systems are simulated with the Monte-Carlo Cluster-Algorithm. During the simulation, the pressure is measured via Equation 11. The measured thermodynamic equation of state is plotted in Figure 27. Each system consists out of $N = 256^2$ hard disks. The pressure only depends on the density η just as in three dimensions. All measurements are performed using periodic boundary conditions, and only the first periodic image is considered. All simulations are carried out with an implementation provided by [6].

To create a system of a certain density, the relation between η and system size L is used. With this equation $\eta = N\pi(\sigma/2)^2 / V = N\pi(\sigma/2)^2 / L^2$ the size can be adjusted to achieve the desired density, where σ is the diameter of a single disk.

As can be seen, the pressure clearly does not follow a monotonic behavior with respect to a change in density. This non-monotonic behavior is evidence for a phase transition, because a change in η does not directly result in a change in pressure.

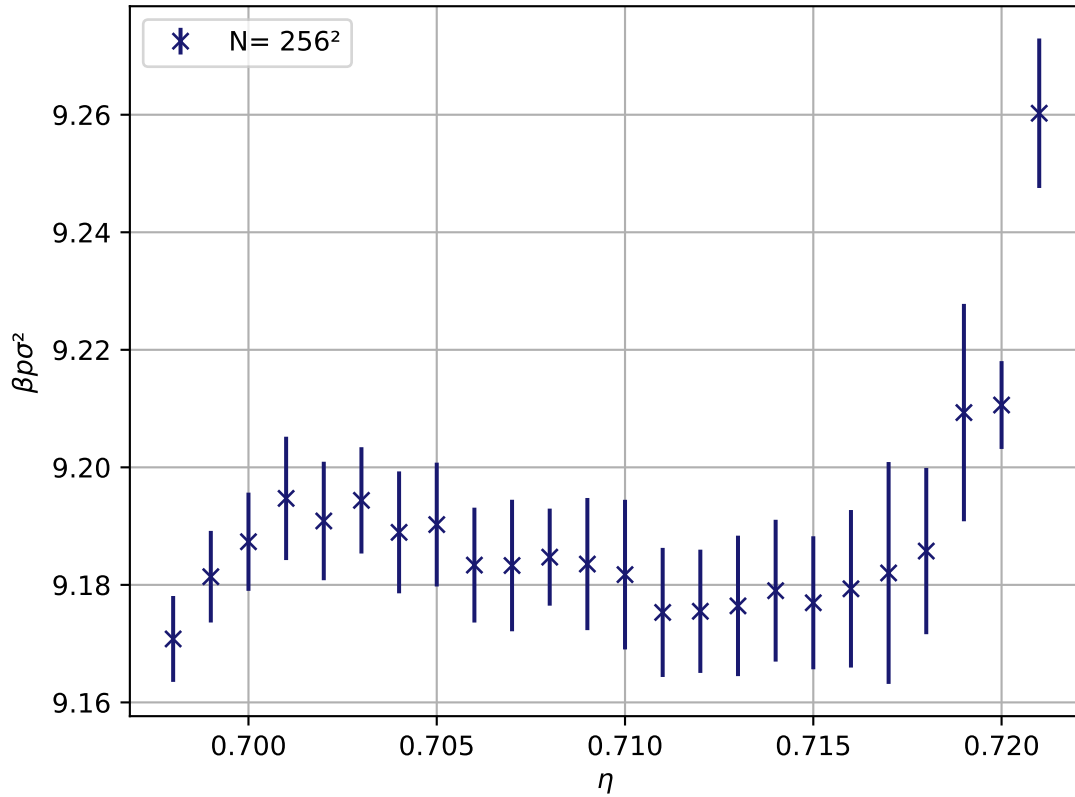


Figure 27: Equilibrium equation of state of a hard disk system with $N = 256^2$ disks. After an initial warm-up period of 50 sweeps, with 10 million Event-Chains each, the pressure is measured during every sweep. The uncertainty is the standard deviation of these measurements. A warm-up of 50 sweeps is chosen because all systems should have equilibrated after this period of time. The equilibration time varies with η . Generally the equilibration time increases in the proximity of a phase transition. This gives an indication that these densities are close to a transition point. Two exception are the system with $\eta = 0.716$ and $\eta = 0.718$, which only equilibrated after about 40 and 30 sweeps respectively. In comparison, no excessive density changes were found in the system with $\eta = 0.717$ after just 10 sweeps. Each system runs 100 sweeps each.

7.2 Orientational order [L. Annuss]

To gain additional insight into the system, the orientation of the disks is studied. The analysis of the orientational order follows largely the step of Krauth *et al.* [22] and the supplementary material. A plain disk has no intrinsic orientation, because no direction is intrinsically favored. Nevertheless, each disk k can be assigned an orientation by taking

into account the position of its neighbors l . The orientation of k is given by

$$\psi_k = \frac{1}{N_k} \sum_l \exp(6i\phi_{l,k}).$$

N_k is the number of neighbors of disk k and $\phi_{l,k}$ is the angle between the connecting line of k to l with respect to a fixed axis.

In this analysis the fixed axis is the unit vector pointing along the x-axis $(1, 0)^\top$. If the disk and its neighbors are placed in a triangular lattice $\psi_k = 1$ a rotation of the plane by $\pi/3$ would not change the configuration with respect to the fixed axis. The orientation of the whole sample is given by

$$\psi = \frac{1}{N} \sum_{k=1}^N \psi_k.$$

As a direction result follows that $\psi = 1$ if all the disks are parallel to the fixed axis and $\psi = -1$ if all the disks are antiparallel.

Two disks k and l are considered neighbors if the midpoint of the line that connects their centers, is closer to k and l than to any other disk j . It is evident that $l \neq k \neq j$ must be fulfilled. Because ψ_k is a complex number, it can also be thought of as a two-dimensional vector $\vec{\psi}_k = (\Re(\psi_k), \Im(\psi_k))^\top$. For better visualization, the spheres are colored according to the projection $\vec{\psi}_k \cdot \hat{e}_\psi$, with $\hat{e}_\psi$ being the unit vector of ψ . An example of a disk configuration with their orientation is given in Figure 28.

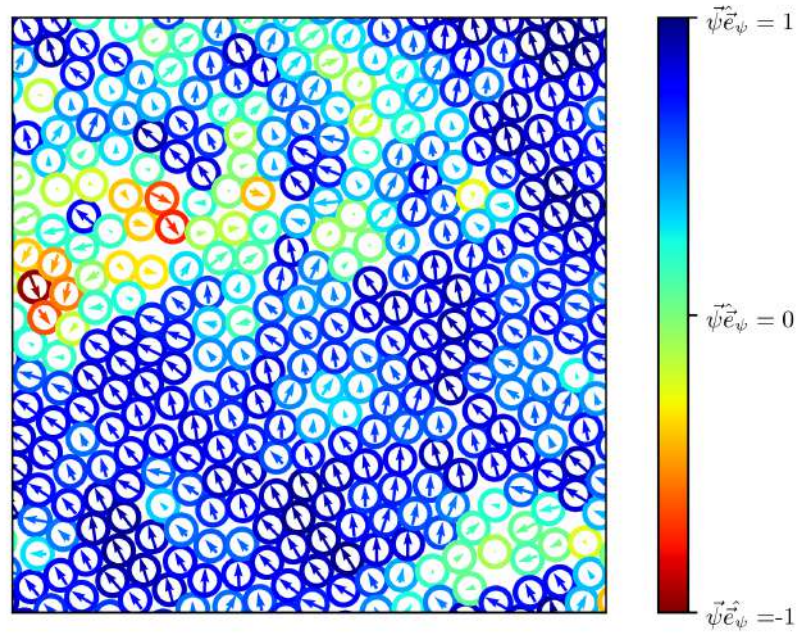


Figure 28: A cut-out of a configuration with $\eta = 0.7185$, where the color of each sphere is determined by the dot product $\vec{\psi}_k \cdot \hat{e}_\psi$. Disks that are colored in blue point parallel to the sample direction, and the red ones in the opposite direction. Green disks point perpendicular to the sample orientation.

In the following, the local orientation of an entire sample is considered, and how this changes with density. Snapshot of systems that lie inside the region where the phase transition is suspected are depicted in Figure 29. Here, it can be seen that during the pressure loop, as can be seen in Figure 27, a majority phase and a minority phase coexist with each other. This is a sign of a first-order phase transition.

Another way to look at the orientational order is the orientational correlation order function $G_6(|\vec{r} - \vec{r}'|)$. This function is the orientational analogy to the pair correlation function $g(r)$ and is calculated via

$$G_6(|\vec{r} - \vec{r}'|) = \langle \psi(\vec{r}) \psi^*(\vec{r}') \rangle,$$

where $\langle \dots \rangle$ implies the averaging of all possible disk combinations that fulfill the positional condition $|\vec{r} - \vec{r}'|$ of the argument. In the following, only the radial symmetric $G_6(r)$ is considered. The different phases should follow the behavior mentioned in Table 5. In a crystal, where the disks form a lattice without any defects, all the disks are aligned in the same direction. This should result in a constant $G_6(r) = 1$. In a solid local defects in the lattice, reduce the order and $G_6(r)$ approaches a constant value at large distances $G_6(r) > 0$. There is an algebraic decrease in the hexatic phase and an exponential decrease in the liquid phase [27].

The orientational order function for the same systems as in Figure 29, with an additional system at $\eta = 0.720$, are shown in Figure 30. As can be seen, the higher the density the higher the orientational order. The behavior for large distances is decisive for distinguishing the phases. Here the three cases from Table 5 are considered, first the expected constant value for high densities is visible, and then there is the transition between the two lowest densities. Where $G_6(r)$ changes from an algebraic to an exponential decay. With this method, the transition point between a solid and a hexatic phase is detected between $\eta = 0.715 - 0.716$ and the hexatic-liquid transition at around $\eta = 0.702 - 0.701$. The two systems above and below the identified transition points can be seen in Figure 31.

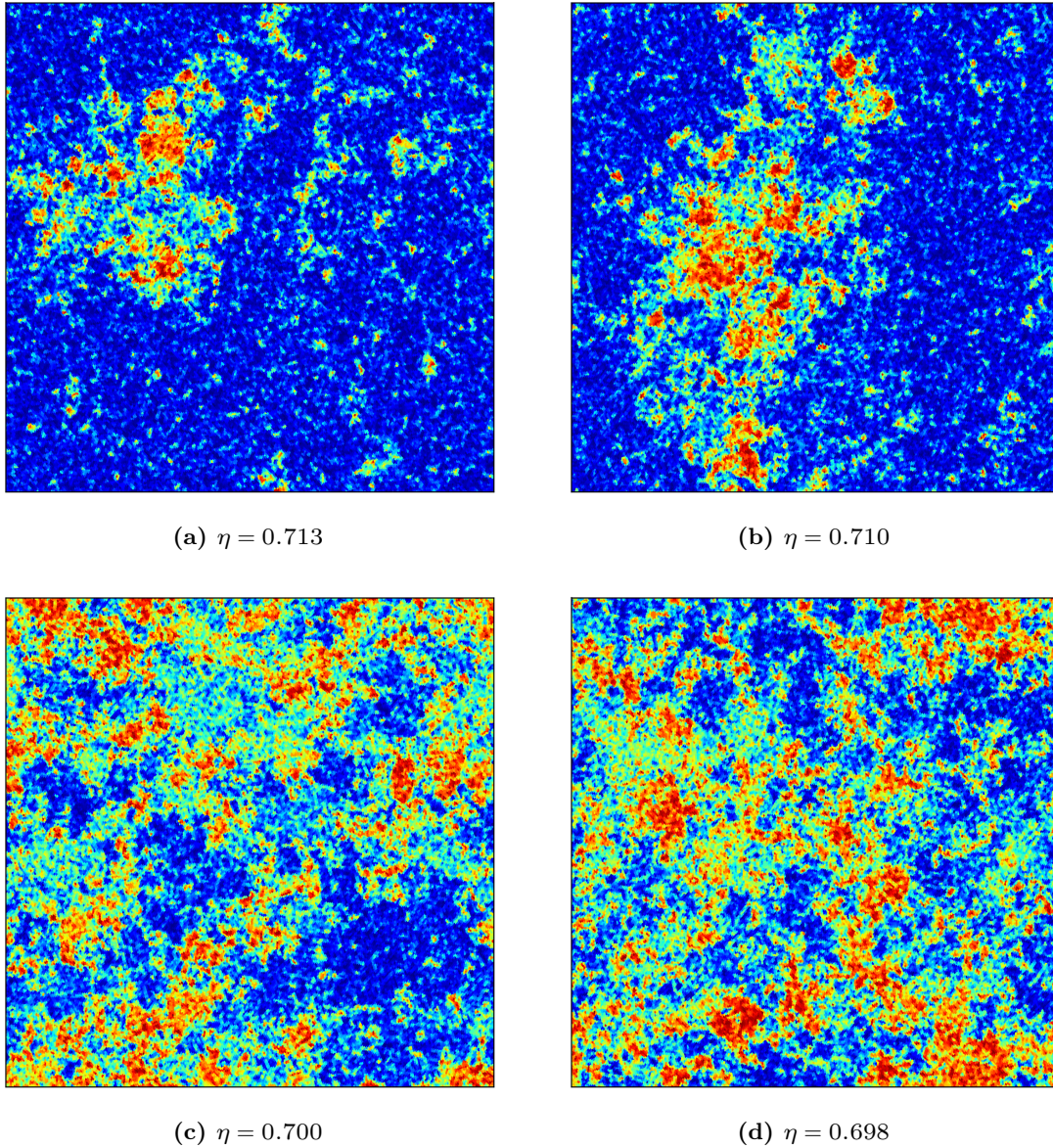


Figure 29: This image series shows the transition between $\eta = 0.715$ and $\eta = 0.700$ regarding the local orientation, blue disks are orientated parallel to the sample orientation and red antiparallel. The orientation of green disks point perpendicular to the sample orientation. The highest density shows that the majority of disks are blue, and only a minority is red. The patches of red disks are all surrounded by green disks. There are also many sections where the blue order is disrupted by only a small number of green disks. At a certain point a bubble of unordered disks forms (a). This bubble configuration expands into a stripe where the ordered and unordered phases coexist (b). At even lower densities, the bubble configuration reverses and the ordered phase forms a bubble inside the unordered (c). This bubble disappears at even lower densities (d).

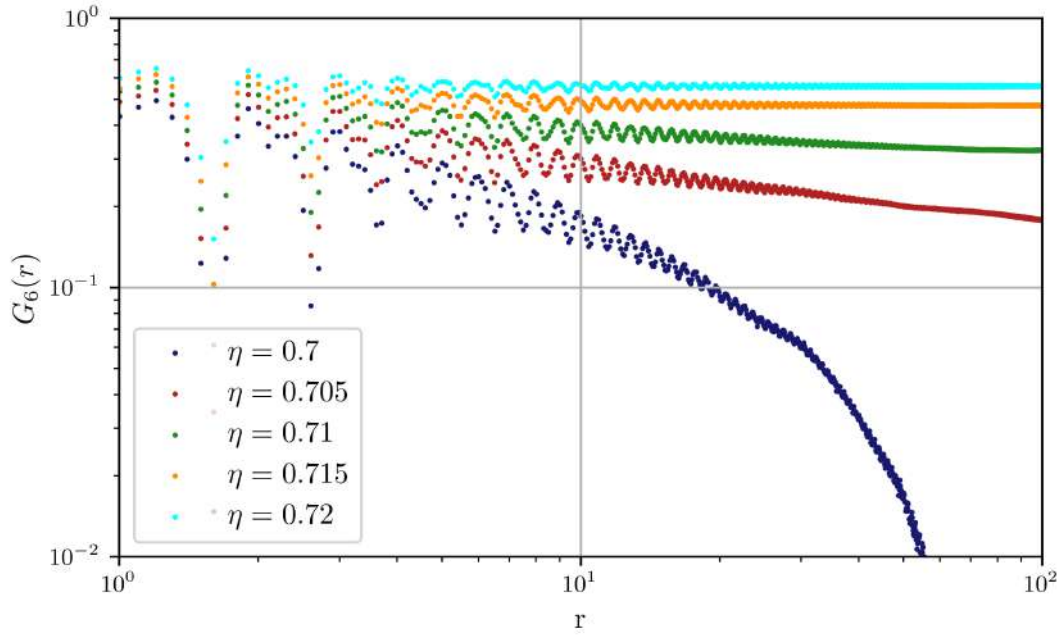


Figure 30: Plotted is the orientational order $G_6(r)$ for five different system densities. The periodic minima occur as a result when a disk lies between two points in the lattice. In general, there is little orientational correlation between two disks with little positional correlation. Note that both axes use logarithmic scales.

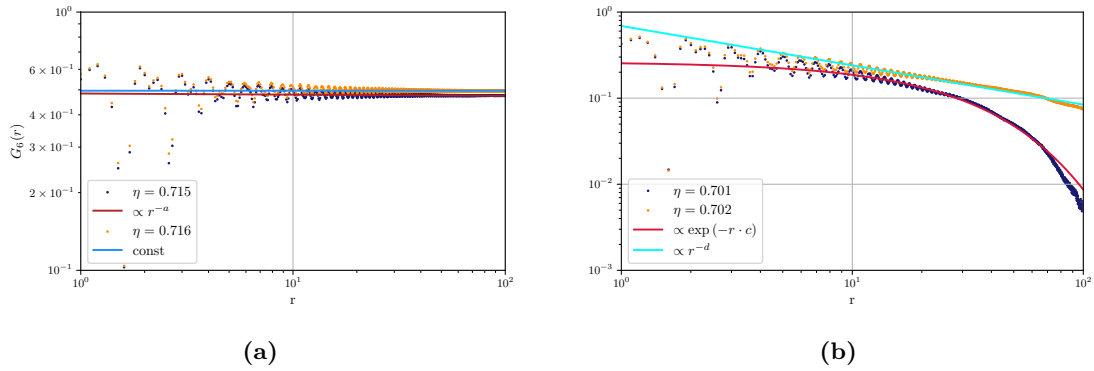


Figure 31: Transition between an algebraic and a constant asymptotic behavior (a) and between the hexatic and liquid phase (b). The fits are for large r .

7.3 Positional order [L. Annuss]

The positional order of a hard disk system can be measured with the pair correlation function. The pair correlation function $g(r)$ measures how much the density varies as a

function of distance r . A more detailed discussion is given in subsection 2.4. Often, only the Euclidean distance r between two particles is used. Because the hard disk system has only two dimensions, the easily computable and displayable $g(x, y)$ can be used. In practice $g(x, y)$ is a normalized histogram of the periodic pair distances $\Delta\vec{r}_{k,l}$ between disks k and l , with

$$\Delta\vec{r}_{k,l} = \vec{r}_k - \vec{r}_l = \begin{pmatrix} x_k \\ y_k \end{pmatrix} - \begin{pmatrix} x_l \\ y_l \end{pmatrix}.$$

Again the distance with $k = l$ is excluded, because this distance does not provide any real information. After this, the histogram is rotated in such a way that the lattice aligns with the x-axis. For this, the whole sample is rotated by

$$\theta = -\frac{1}{6}\phi_\psi + \frac{\pi}{6},$$

where ϕ_ψ is the angle between ψ and the x-axis. The histogram is rotated via the *scipy.ndimage.rotate* function [28]. Examples for two different densities are given in Figure 32.

With the system properly aligned, a slice of the histogram at $g(x > 0, 0)$ is taken. The histogram is discretized with a bin size of $\sigma/10$ in x and y direction. The change of the positional decay from hexatic to short-range can be seen at higher distances. The transition between the solid and hexatic phase could be established to be between $\eta = 0.718$ and $\eta = 0.719$.

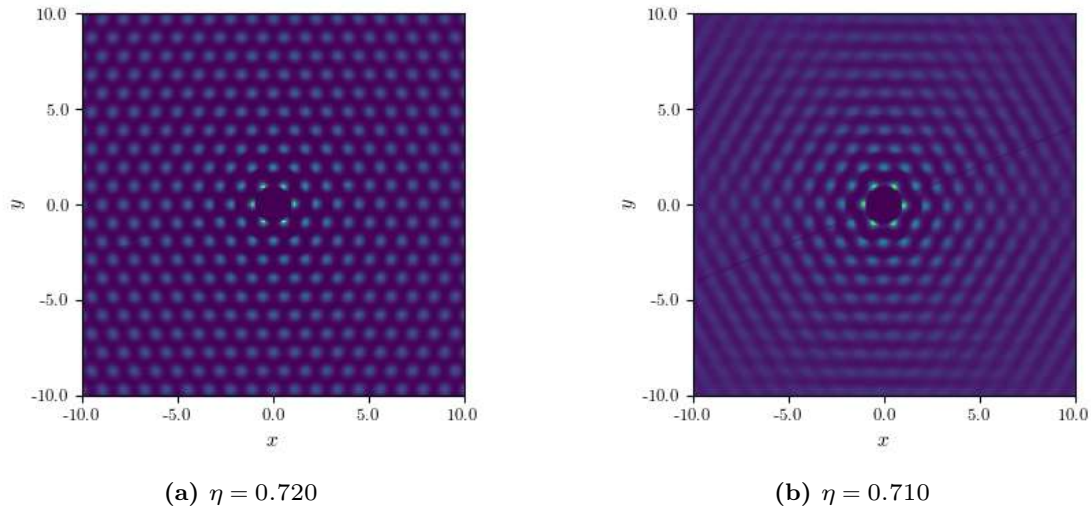


Figure 32: The two-dimensional histograms for different densities with $\eta = 0.721$ (a) and $\eta = 0.710$ (b). Yellow bins contain a high number of disks, and blue bins a low number. Peaks in the high-density plot are much more localized than in the low-density plot. As expected, the nearest neighbors are the most localized. Both histograms show the exclusion area around disk k . The diagonal line from the lower left to the upper right is an artifact from the binning process and should not significantly influence the further results.

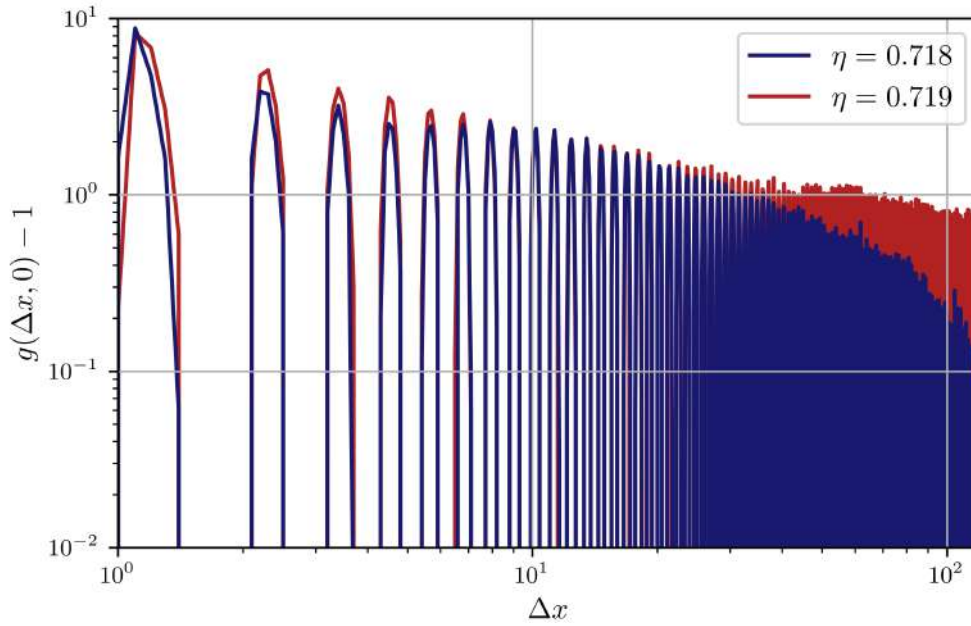


Figure 33: Shown are the normalized slices of the two-dimensional histogram for the system above and below the transition point. Note the double logarithmic axis. The y-axis is subtracted by -1 to highlight the decay against zero for large Δx .

7.4 Summary and discussion [L. Annuss]

First, the equation of state was measured.

The curve follows largely the reference values given by [22]. The pressure clearly exhibits non-monotonic behavior, which is clear evidence of a transition. Additionally, the uncertainties in the measured values are larger within the transition region than outside it. This fits to the expectation that the pressure is subject to higher fluctuations near a phase transition.

The significantly larger uncertainties are probably a consequence of the low sample size of only 50 measurements per system. Each Event-Chain has a length of 20. Together with an approximated mean free path of ~ 0.069 each chain displaces ~ 290 disks. This should result in a behavior where the simulation operates much faster than a local Monte-Carlo algorithm, and that the entire system is not translated with a single chain. During every measurement a disk is displaced ~ 44000 times. This should ensure that each measurement is independent.

The transition depicted in Figure 29 shows the transition from an ordered to an unordered phase. When analyzing systems with the same densities as in Figure 29, a bubble or stripe configuration could not always be detected. Often, two smaller bubbles or a multitude of

spots from the minority phase can be seen within the majority phase. One explanation is that the simulation time is too short and that the system did not not enough time to completely phase separate.

The analysis of $G_6(r)$ shows the Kosterlitz-Thouless solid-hexatic transition point at $\eta = 0.715 - 0.716$ and the hexatic-liquid transition at $\eta = 0.702 - 0.701$. The first value deviates from the other studies significantly. There, the transition was determined to be at $\sim \eta = 0.72$ [22]. The estimation of the transition point could be improved by an increase in the system size. Because the accuracy is limited by the size of the system. A more intrinsic problem with using $G_6(r)$ in case of the solid-hexatic transition is that $\lim_{b \rightarrow +0} c \cdot r^{-b} = c$ and the change from $b = 0$ to $0 < b < \epsilon$ is not practically detectable. The second value $\eta = 0.702 - 0.701$ is closer to ~ 0.70 given by [22].

The transition from solid to hexatic is also measured via the positional order and yields a range of $\eta = 0.718 - 0.719$. This lies significantly closer to the reference value than the value given by the orientational order function. Here, several sources of error need to be mentioned. First, there is the discretization error, which occurs when the histogram is created, and second, the interpolation error, which arises when the histogram is rotated. The interpolation error is a result of trying to rotate the discrete histogram by a continuous angle. The error should be negligible because of the high order of interpolation used.

At this point, several avenues for further analysis emerge that could provide a deeper understanding of the system's behavior. One key area are finite-size effects. In further analysis, it should be investigated where and how much finite-size effects have to be considered when determining the exact point of the transition. Another essential point is to better quantify the transition by establishing uncertainties of the transition point. Averaging the transition points over multiple simulations with varied initial configurations would help refine these estimates, because in this study only individual systems have been analyzed. Lastly, the local density variations could be examined. This would provide a complementary feature throw in which the coexisting phases could be distinguished.

8 Sedimentation of Hard Spheres [R.-M. Lehner]

On Earth, all experimental and theoretical systems are subject to gravity. While this force can be neglected on many microscopic scales, there are still relevant and interesting phenomena that can only be observed due to gravity. Since the hard-sphere system simulates both solids and gases/liquids, systems exhibiting a coexistence of both phases can be investigated here.

When gravity is introduced into such a system, sedimentation might occur; depending on the system parameters. This means that the gravitational force translates into a pressure that depends on height. If the pressure is high enough, a height-dependent phase transition occurs and a solid phase forms. Consequently, the density must also depend on height. Height and pressure, as well as height and density, are directly related, so only one of the two needs to be measured.

8.1 Introduction and Preparation

This section is divided into two parts: the first part outlines the fundamentals and conditions under which sedimentation was simulated, while the second part presents and analyzes the simulation results.

8.1.1 Validation of the Simulation

Since a different implementation was used for this part of the project compared to the rest, the physical correctness is verified using the previously introduced Carnahan-Starling approximation from Equation (3). Figure 34 and Figure 35 show that the measured values correspond to the desired behavior. Small deviations toward higher pressures at higher densities were observed. These deviations can be explained by the method used to populate the grid and the total equilibration time. Simple primitive cubic (cp) lattices with random perturbations are used to populate the simulation which results in high entropy states. Since simple pc lattices allow for a packing fraction η of only about 0.524, an alternative initialization method was necessary. For densities higher than 0.524, the spheres were arranged in a hexagonal close-packed (HCP) lattice, which enables fractions up to approximately 0.74. While this allows for higher densities, longer equilibration times are likely required to reach equilibrium. Despite this, a clear phase transition is visible. Furthermore, the radial distribution function $g(r)$ indicates that there are no overlaps between spheres. The observed behavior also matches the chosen density of $\eta = 0.6$; this behaviour was previously discussed in Figure 6.

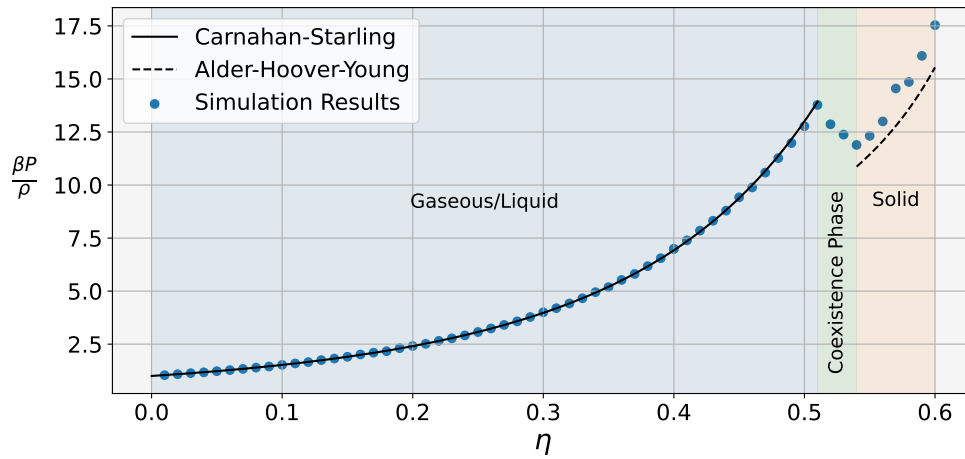


Figure 34: Pressure-density relationship for a hard-sphere system. The x-axis is divided into three phase regions: gas/liquid, coexistence, and solid, indicated by different colors. The simulation results (blue dots) show a clear phase transition. The Alder-Hoover-Young approximation (dashed line) is included for comparison in the solid phase region [29]. A slight overestimation of pressure is observed in the solid phase. Simulations were performed with a box size of 20σ , a displacement length of 2σ , and 10 million chains.

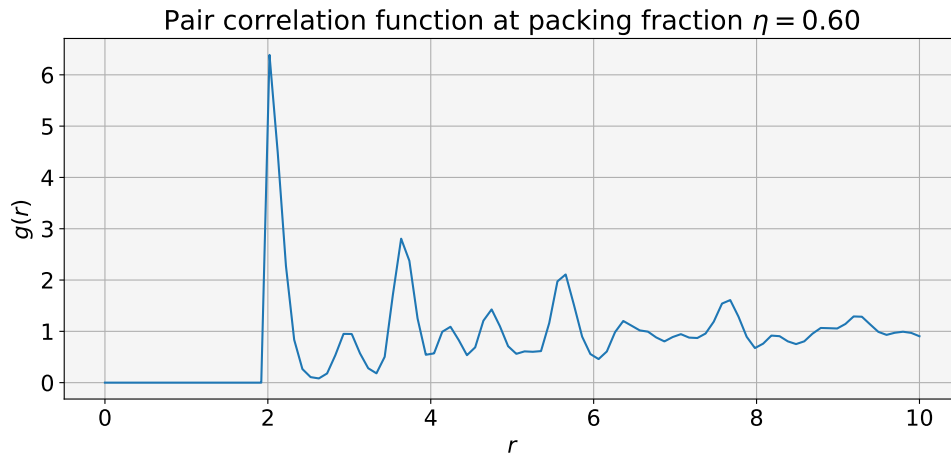


Figure 35: Radial distribution function $g(r)$ for a hard-sphere system at a packing fraction of $\eta = 0.6$. The function is zero for $r < \sigma = 2$, as expected due to the hard-sphere diameter. Small peaks are visible beyond $r = 2$, indicating the formation of a solid structure. The simulation used 9,872 particles, 100 bins, a displacement length of 2σ , and 10 million chains.

8.1.2 Modification of the Simulation

To introduce gravity, two modifications must be made:

1. Both the floor and the ceiling must be fixed. The walls in the x and y directions can remain periodic.
2. Gravity must be introduced. Care must be taken to ensure that the simulation remains physically accurate. This means, for example, that the barometric height formula is satisfied and energy is conserved.

In both cases, an additional lifting event is triggered. This means that both the fixed walls and the gravitational potential generate a virtual collision. Both have in common that the direction of the chain is mirrored in the Z-direction.

8.1.3 Fixed Walls

A fixed wall can be regarded as an infinitely high potential that the sphere cannot overcome:

$$V(z) = \begin{cases} \infty & \text{if } z < 0 \text{ or } z > L \\ 0 & \text{otherwise} \end{cases} , \quad (36)$$

With L being the height of the simulation volume. This approach allows for consistent handling of potentials in the implementation—such as the gravitational potential. However, in this case, potentials were not used for fixed walls; instead, only the distance to the ceiling or floor was calculated. If the sphere were to come closer to the wall than a radius r , a lifting event would be triggered. This behavior is illustrated in Figure 36.

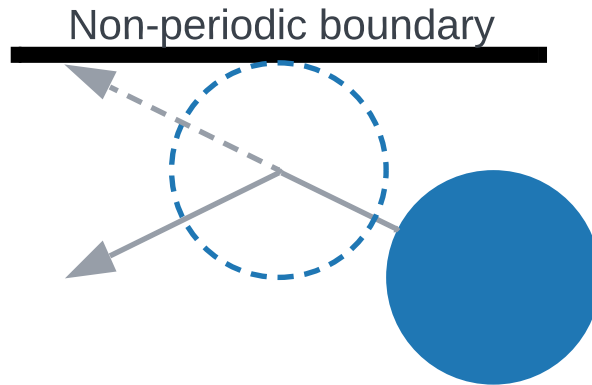


Figure 36: Collisions with a non-periodic boundary are added as a lifting event with the line parameter t to the list of potential collisions, provided the sphere can approach the wall closer than its radius r . If this is the first collision to occur, the chain direction along this axis is inverted.

8.1.4 Gravity

To implement gravity, the gravitational potential is used, which determines how costly movement in the positive z -direction is. Therefore, when a particle moves in the positive z -direction, it consumes energy. For each movement, a random energy budget is determined that the particle is allowed to expend. This results in a specific distance the particle may move. If this distance lies within the chain length, a lifting event is added, which is interpreted as a virtual collision. If no other particle is found as a collision partner before this virtual event, the virtual collision occurs. Here too, the particle is reflected in the z -direction, and the chain continues downward.

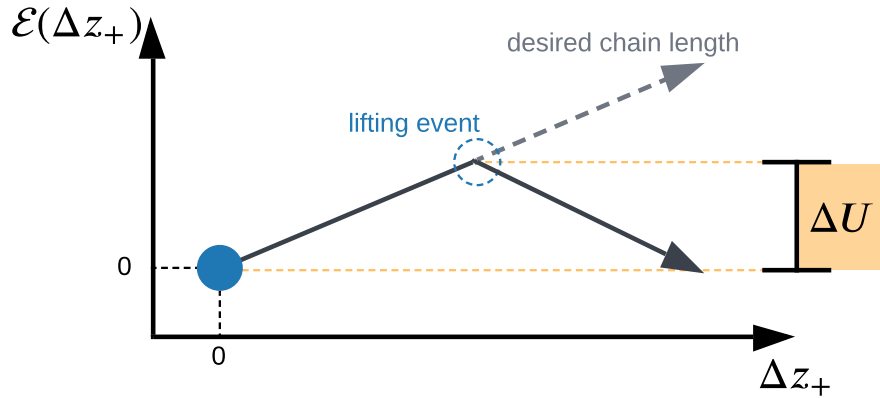


Figure 37: Concept of the gravitational potential that triggers a lifting event once the energy ΔU is expended. Δz_+ is the distance traveled by the sphere in the positive z -direction. The energy $\mathcal{E}$ corresponds to the potential energy mgh , where $h = \Delta z_+$.

Figure 37 schematically illustrates how such a potential functions within the simulation. The energy required for a move is determined by the potential energy

$$\mathcal{E}(\Delta z_+) = mg\Delta z_+ \quad , \quad (37)$$

where $m = V_p \Delta \rho$ is the reduced mass of the particle (with V_p as particle volume and $\Delta \rho$ as difference between the densities of particle and solvent [30]), g is the gravitational acceleration, and Δz_+ is the distance traveled in the positive z -direction. Following the work of Kierfeld et al., the energy budget is determined randomly [3], with

$$-\ln u = \Delta U = \int_0^{\Delta z_+} dz [\partial_z \mathcal{E}(z)]^+ \quad , \quad (38)$$

where u is a uniformly distributed random number between 0 and 1 and Δz_+ is the distance traveled in the positive z -direction. $\mathcal{E}(z)$ is the potential energy at height z , which simplifies the total energy budget to $mg\Delta z_+$ in this case.

The energy budget is then compared to the energy required to move the particle. If the energy budget is sufficient, the particle is moved, and no lifting event is triggered. If the energy budget is exceeded, a lifting event is added similar to the fixed wall case.

In Figure 38, four scenarios are depicted that can occur during movement in the positive z -direction.

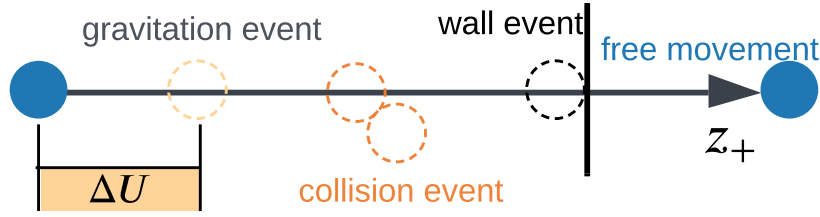


Figure 38: Four possible events that can occur during movement in the positive z -direction: (1) gravitational event, where the particle used up its energy budget; (2) collision event with another particle; (3) collision event with a fixed wall; and (4) no event, where the particle can move freely.

This approach of drawing a random energy and triggering lifting events preserves the benefits of the event-driven nature of ECMC—namely, that there are no rejected moves. Every move is permitted, and once the energy is expended, only the direction of the chain is reversed.

8.1.5 Gravitational Length

The strength of gravity can be freely chosen using the factor mg . In the case of gravity, a new characteristic length scale of the system emerges—the gravitational length:

$$l_g = \frac{k_B T}{mg} . \quad (39)$$

This length is the scale over which changes in pressure might occur. Therefore, the smaller l_g is, the more pronounced the pressure gradient will be. This becomes evident when either scaling the gravitational potential or the temperature. A higher temperature results in a higher pressure which drives the particles apart; a higher gravitational potential results in a stronger force pulling the particles downward and together at the bottom. The gravitational length is a measure of the interplay between these two forces.

8.2 Simulation Results

All simulations were conducted with the parameter $l_g = 1$, which represents a good compromise between system size, particle number, sedimentation behavior, and computational runtime. Figure 39 shows the simulation results for $N = 91125$ particles, 100 million chains, and a box size of 50σ . It is evident that the lower layers form a solid structure, while a phase transition occurs in the middle layers. This phase transition takes place over approximately three layers, or 3σ . A structural analysis using the Ackland-Jones method [31] assigns colors to particles based on their crystal structure. Gray particles cannot be assigned to any specific structure, while red particles exhibit

hexagonal close-packed (HCP) arrangements. Green particles indicate an FCC crystal structure. However, the lower solid layers do not form a perfect solid. Defects are visible, as well as transitions from HCP to FCC. Some defects are also present at the periodic boundaries. The bottommost layer cannot be assigned a structure because potential neighbors are missing.

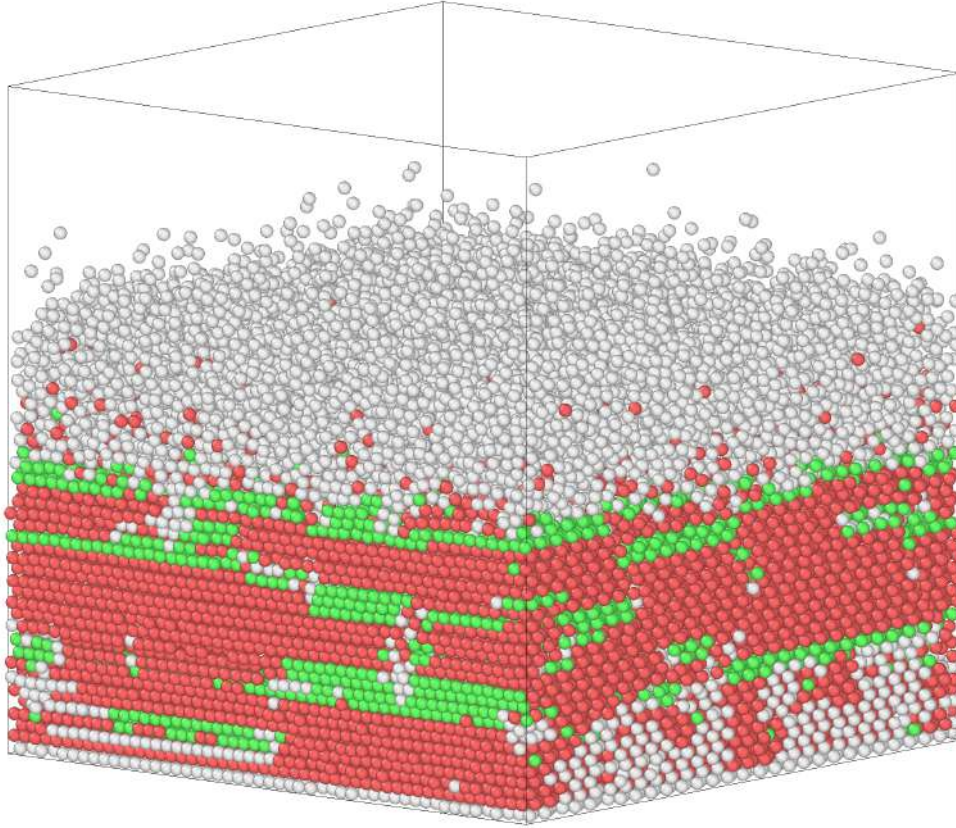


Figure 39: Simulation results for $N = 91125$ particles, 100 million chains, and a box size of 50σ . The lower layers form a solid structure with a clear phase transition in the middle layers. Particle colors indicate structures: **gray** (undefined), **green** (FCC), and **red** (HCP). Defects and HCP-to-FCC transitions are visible in the lower layers.

8.2.1 Height Profiles in Pressure and Density

The next step is to investigate how the density and pressure depend on the height; and if we can reproduce the Carnahan-Starling approximation. To determine the height-dependent packing fraction η , scans along the z -direction are performed on distinct layers, as illustrated schematically in Figure 40. A layer height of approximately 3σ is chosen, as this corresponds to about three layers and allows for the detection of changes within these layers. It is also important to consider a sufficiently thick layer because a layer that is too narrow causes periodic oscillations in the density due to the solid structure.

However, if the layer is too large, the density curve becomes too smooth and fails to detect a phase transition.

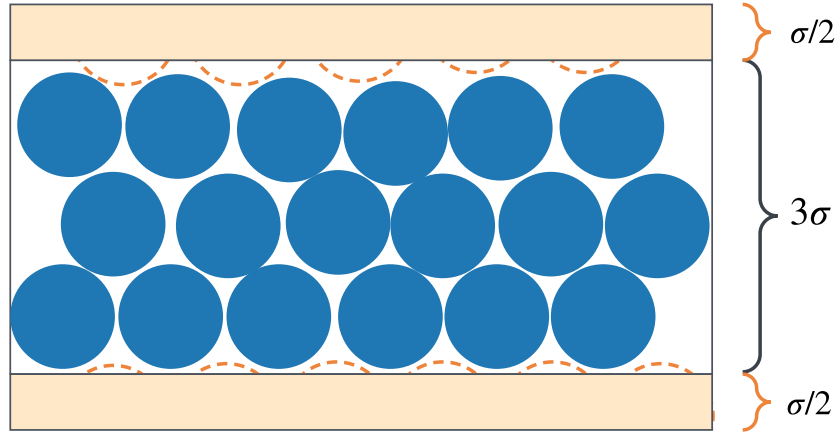


Figure 40: Schematic illustration of a 3σ -wide layer of spheres with boundary layers of $\sigma/2$ on the top and bottom. These boundary layers account for the volume of spheres partially overlapping with the central layer.

Figure 41 presents a measurement using the z-layer scans. As expected, the density is height-dependent. Even though fluctuations in the density are noticeable, a phase transition is not directly apparent. This may be due, on the one hand, to the logarithmic scale, but on the other hand, it could also be attributed to the method used for the z-scan. The scan values used here are those that, after some trial and error, resulted in the least density fluctuations at high densities. However, it is conceivable that a different choice of scan values could lead to a visible phase jump.

Significantly better, however, is the pressure measurement in Figure 42, which provides an astonishingly good approximation to the previously considered values. Although the reference values are generally slightly underestimated, such inaccuracies are to be expected with such a coarse measurement. The phase transition is less pronounced but still recognizable. It should be emphasized that the pressure measurement here is carried out indirectly via the density. Piazza et al. have shown that the pressure in such a system can be determined directly from the density [30]:

$$\frac{\Pi(z)}{k_B T} = l_g^{-1} \int_z^h n(z') dz' \quad . \quad (40)$$

Here, $n(z)$ is the normalized particle density, l_g is the gravitational length introduced earlier, h is the height of the system, and $\Pi(z)$ is the hydrostatic pressure. Therefore, the pressure measurement is an indirect measurement of the density, which nevertheless

shows good agreement with the reference values. It is important to note that a reference pressure P_0 must be added to the hydrostatic pressure to determine the pressure in a system with gravity. This reference pressure is the pressure that would exist in a system without gravity above the height h . In a system in equilibrium, the following applies:

$$\frac{dP(z)}{dz} = -n(z)mg \quad , \quad (41)$$

and integrating this equation leads to

$$P(h) - P(z) = \int_z^h n(z') mg dz' \quad . \quad (42)$$

From this, the two equations follow:

$$P(z) = P_0 + \int_z^h n(z') mg dz' \quad , \quad P(z) = P_0 + mg\Pi(z) \quad . \quad (43)$$

In order to achieve agreement with the Carnahan-Starling approximation, the reference pressure P_0 must be chosen such that the measured pressures match the reference values, from which P_0 is determined to be 1.

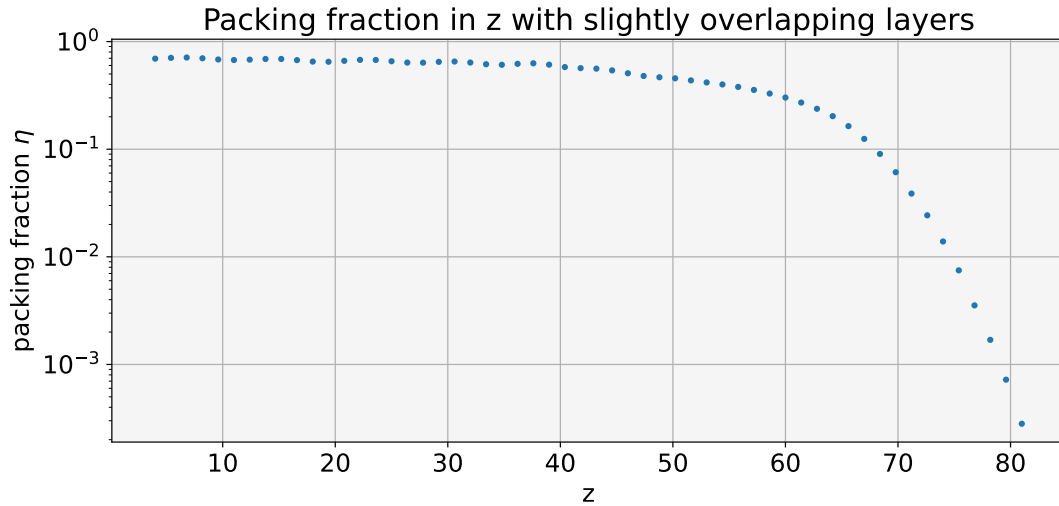


Figure 41: Height-dependent packing fraction η obtained from z -layer scans. Fluctuations are noticeable, but a phase transition is not directly apparent. The overlapping factor is set to 0.3σ .

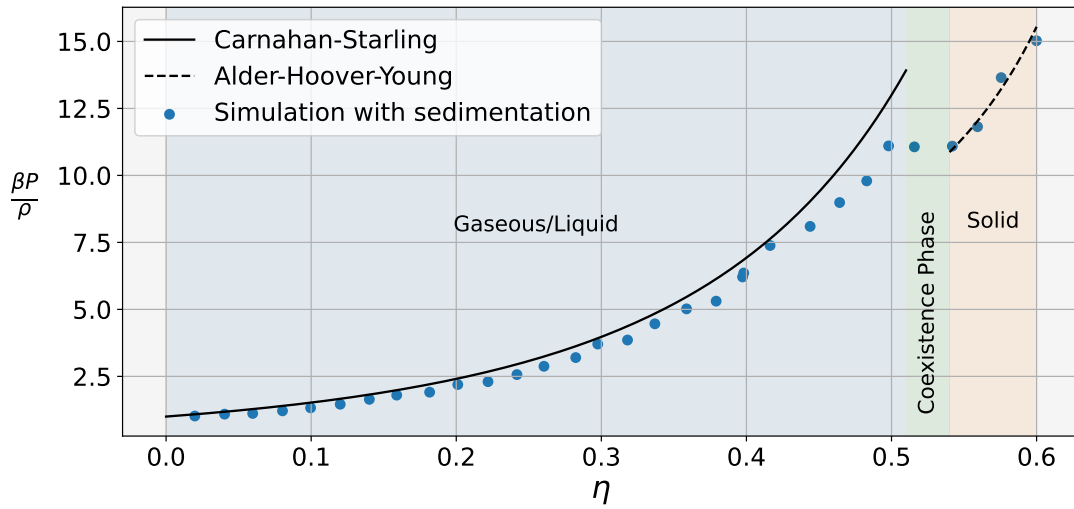


Figure 42: Pressure as a function of packing fraction η . The pressure measurement provides a good approximation to the reference values, with the phase transition being less pronounced but still recognizable.

In theory, it should be possible to remove lower layers of spheres without affecting the dynamics of the layers above. Therefore, when considering the same simulation volume with a reduced number of particles, the shape of the curve should only shift along the x-axis (or in the z-coordinate with respect to the simulation).

Figure 43 confirms this behavior for five different particle numbers.

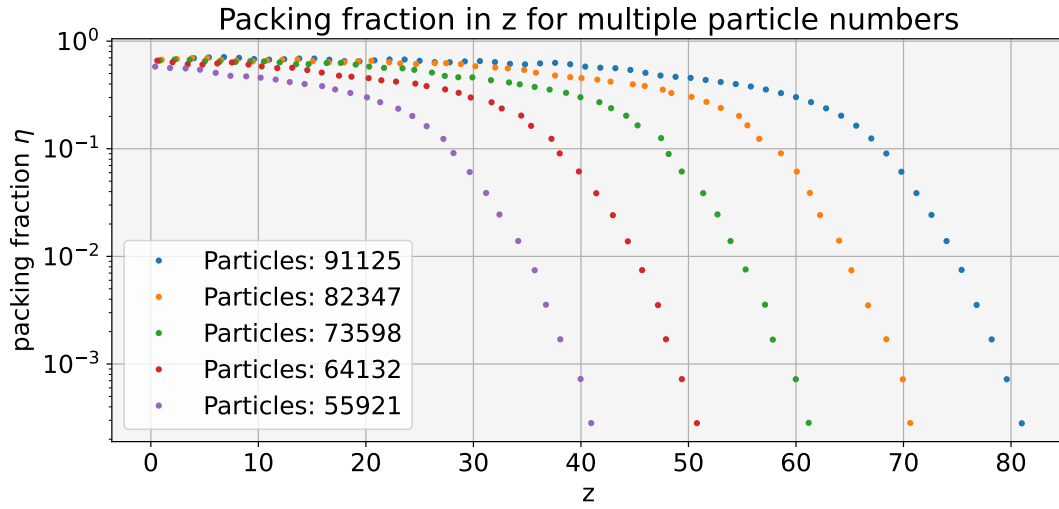


Figure 43: Packing fraction η as a function of height z for multiple particle numbers. The qualitative shape of the curves remains unchanged as the number of particles decreases. Each curve is shifted along the z -axis, confirming that the removal of lower layers does not affect the dynamics of the upper layers.

8.2.2 Cross-Sections

Next, the sedimentation and structure of the lower layers are observed in more detail. For this purpose, cross-sections through the simulation are considered. Side views, in particular, offer valuable insight into the structure of the lower layers. To obtain a comprehensive overview, all particles in the simulation are visualized simultaneously, resulting in a complete picture rather than a localized section. To maintain clarity, the particles are rendered smaller and semi-transparent. The color scheme from the structural analysis is retained for consistency. The simulation considered here corresponds to the one shown in Figure 39.

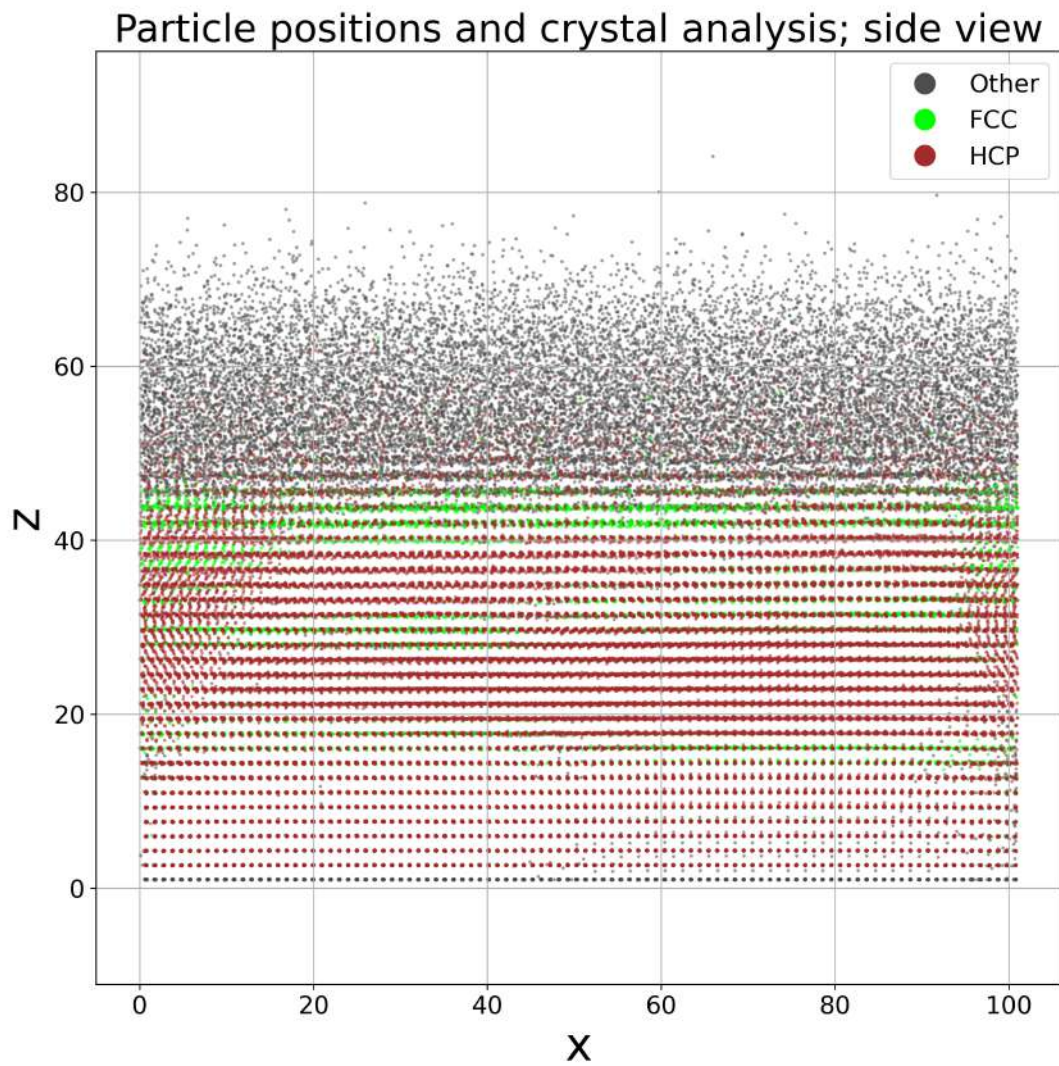


Figure 44: Side view of the sedimentation structure of Figure 39 along the xz -plane. Swirling patterns appear on the left and right sides.

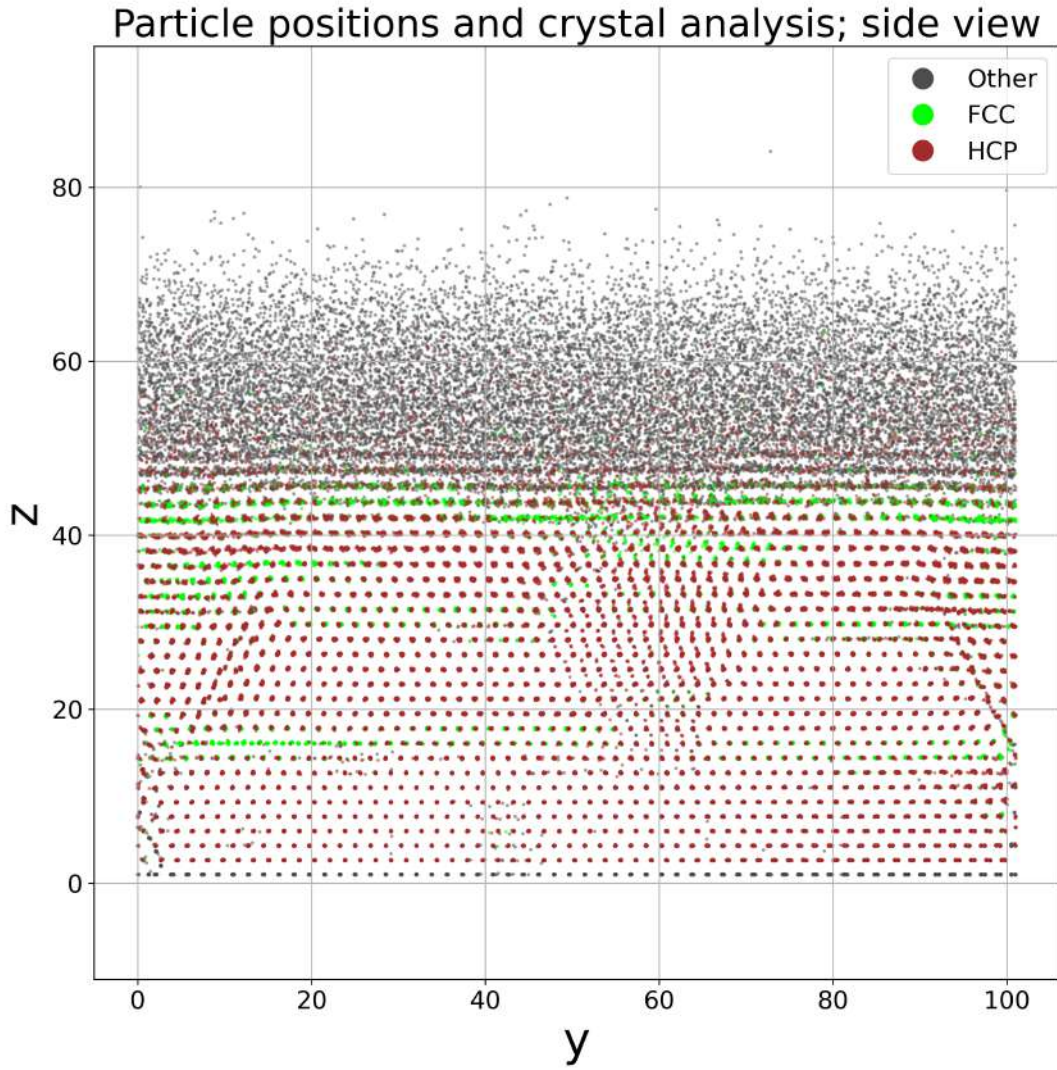


Figure 45: Side view of the sedimentation structure of Figure 39 along the yz -plane. On the left and right sides, some layers appear to be sliding or shearing.

The structures of the lower layers are clearly visible in the side views presented in Figure 44 and Figure 45. Particularly notable is the formation of the FCC crystal lattice, which is expected to emerge with longer simulation times [12]. Defects and transitions from HCP to FCC structures are fairly distinguishable. Additionally, some swirling patterns indicate that the solid structure has not yet fully developed, leading to shearing or even sliding between the layers. The phase transition occurs within only a few layers, as seen in the compact region where the density changes sharply. The bottommost layers cannot be distinctly classified into a specific structure due to missing neighboring particles.

8.2.3 Time Evolution

All measurements considered so far are systems that have been equilibrated for some time (over 100 chains per particle). It remains to investigate the timescale over which sedimentation occurs.

Motivated by the work of Marechal et al., Figure 46 shows the lowest layers at different time intervals [32]. The observed spheres are located at heights ranging from $z = 0$ to $z = 6.5$, and only particles identified as part of HCP or FCC crystal structures are displayed. It is clearly visible how small crystalline zones initially form and then steadily grow. The edges of these zones either merge or form boundary layers. It can also be observed that the layers stabilize after an initial rapid phase and change only slowly afterwards. The system under consideration contains $N = 64000$ particles and was executed over 100 million chains, which corresponds to 15.62 chains per particle per frame. A total of 100 frames are considered (only 79 for the bottom view), recorded at intervals of 1,000,000 chains.

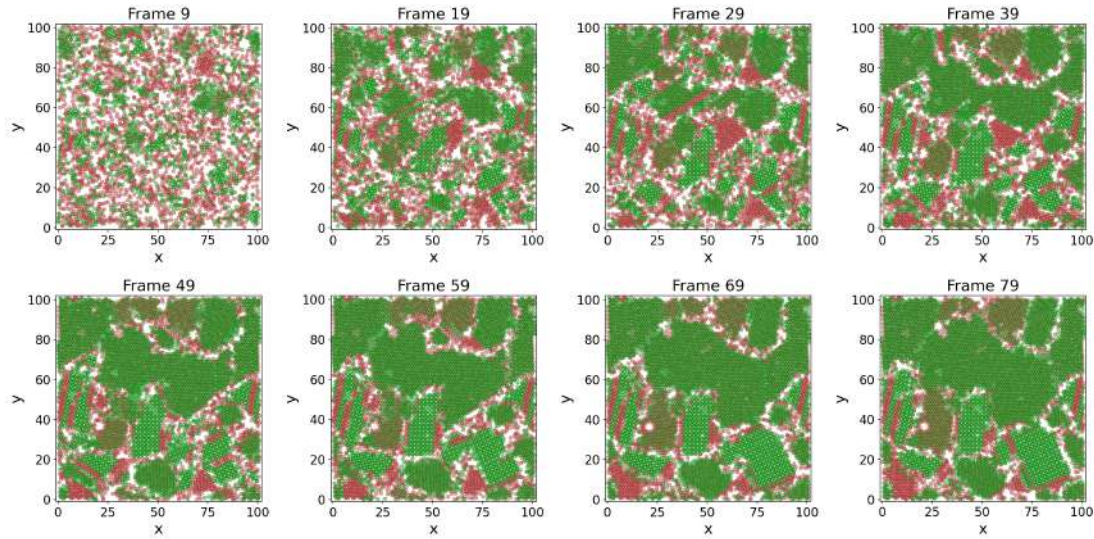


Figure 46: Bottom view of the lowest layers showing FCC and HCP structures for selected frames at different time intervals. Only particles in the height range of $0 \leq z \leq 6.5$ are displayed.

Complementary to the observation of the lowest layers, the entire system and the evolution of HCP, FCC, and unstructured particles are considered. Figure 47 shows the temporal evolution of sedimentation and crystal formation. The number of HCP structures increases rapidly within the first 10 frames, corresponding to approximately 150 chains per particle. After about 20 frames, the number of HCP structures reaches a plateau that is maintained throughout. The number of FCC structures, however, continues to grow almost linearly. The continuous decrease in the number of unstructured particles reflects the ongoing sedimentation and crystal formation process. After approximately 90 frames (about 1400

chains per particle), there are more FCC structures than HCP structures present in the system.

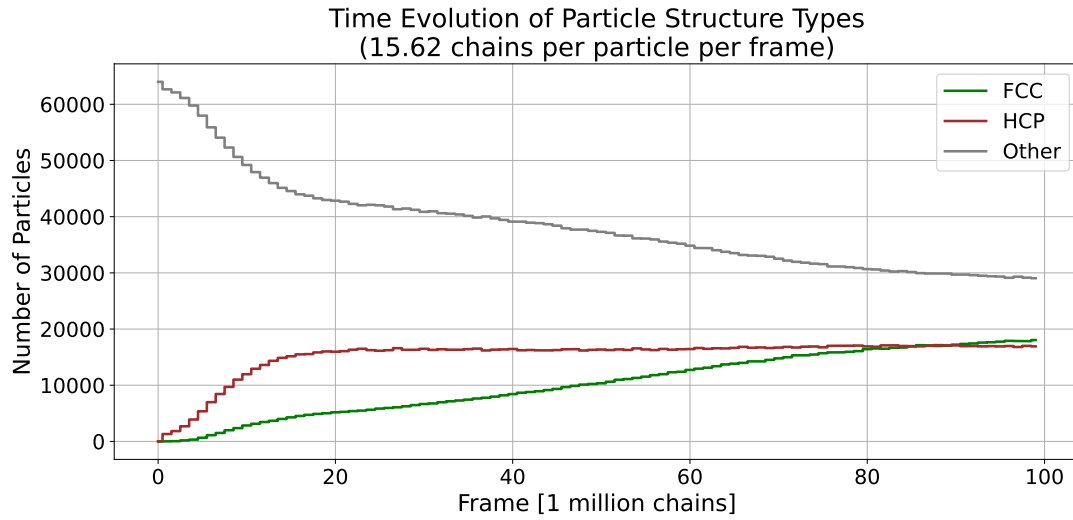


Figure 47: Time evolution of the number of HCP, FCC, and unstructured particles over 100 frames. The number of HCP structures rapidly increases at the beginning and then stabilizes, while FCC structures continue to grow over time.

Finally, the 3D representation of the system is shown in Figure 48. The snapshots are taken at various mentioned time points. The rapid initial increase in HCP structures can be explained by the fact that many particles initially form local crystalline structures, which quickly disintegrate and appear to be "floating" in space. However, the HCP plateau observed at frame 20 suggests that larger structures are forming in the lower part of the simulation. Over time, sedimentation and crystal formation become more evident. Layers of FCC and HCP structures develop, which are not only aligned in the xy -plane but also have extensions in the z -direction, causing them to appear tilted.

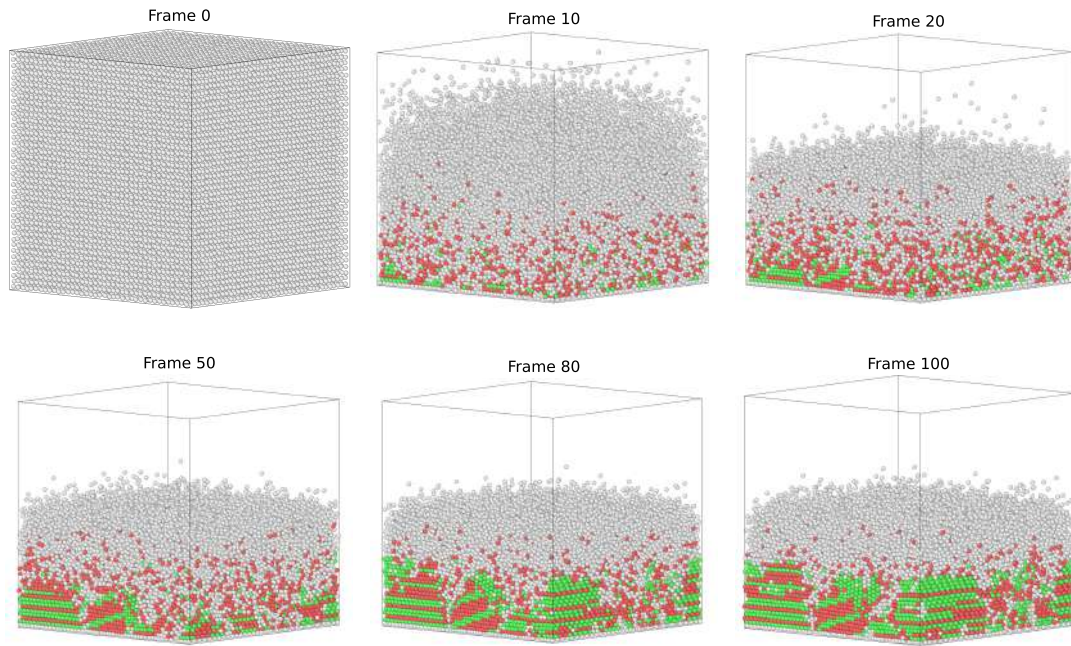


Figure 48: 3D visualization of the system at different time points, highlighting the evolution of FCC and HCP structures.

8.3 Discussion

8.3.1 Summary of Findings

The sedimentation behavior of hard spheres under the influence of gravity was investigated using event-chain Monte Carlo (ECMC) simulations. The introduction of gravity and fixed walls in the z -direction led to the formation of a solid structure in the lower layers and a phase transition in the middle layers, as shown in Figure 39. The Ackland-Jones structural analysis identified both FCC and HCP crystal structures, as well as defects and transitions between these phases.

The height-dependent packing fraction $\eta(z)$ was determined using z -layer scans (Figure 40). Figure 41 confirmed that the density varies with height due to the gravitational field. Although density fluctuations were observed, the phase transition was more evident in the pressure as a function of packing fraction (Figure 42), which aligned better with the Carnahan-Starling equation.

The response of the system to varying particle numbers was also explored. Figure 43 showed that removing lower layers by reducing particle numbers did not alter the dynamics of the upper layers, as the shape of the density curves remained consistent but shifted along the z -axis.

Time-dependent analysis of the sedimentation and crystal formation (Figures 46 and 48)

revealed that small crystalline zones initially formed and grew, with HCP structures stabilizing early and FCC structures continuing to grow over time (Figure 47). This evolution highlights the transition from disordered to ordered phases under gravity.

8.3.2 Limitations and Improvements

Several aspects of the simulation can be improved. The measurements, especially those related to time evolution, were constrained by simulation duration. Longer runs are needed to capture the full development of the solid phase and reach equilibrium, particularly for FCC structures, which continued to grow until the end.

The density determination using z -layer scans showed fluctuations that made the phase transition difficult to observe directly. This issue could be resolved using kernel density estimation to smooth the profile without losing resolution. An optimal balance between smoothing and resolution is needed to detect phase transitions accurately.

The pressure measurements were slightly lower than the values predicted by the Carnahan-Starling approximation, likely due to limited equilibration time. Extending simulation times and considering average collisions per chain could improve accuracy.

Structural analyses using the Ackland-Jones method provided valuable insights but were limited to specific snapshots. More comprehensive, time-dependent structural analyses would offer a deeper understanding of the phase transition and sedimentation processes.

8.3.3 Future Work and Outlook

Extending simulation durations would allow for a more complete observation of sedimentation and system equilibration. This is especially important for capturing the formation of FCC structures and understanding long-term dynamics under gravity.

Investigating systems of varying sizes would provide insight into how size affects sedimentation dynamics and phase behavior. Both small and large systems could uncover scaling relationships and finite-size effects influencing crystalline structure formation.

Additionally, conducting more detailed time-dependent crystal structure analyses would help understand the nucleation and growth processes of different phases. This would clarify the conditions favoring FCC over HCP structures and the role of defects and grain boundaries in sedimentation.

Exploring the impact of different gravitational lengths l_g could provide insights into the balance between thermal motion and gravitational forces, enriching the understanding of sedimentation in hard-sphere systems.

In conclusion, while the current study offers valuable insights into sedimentation under gravity, there are many opportunities for further refinement and exploration. Addressing the identified limitations and working on the suggested directions could help to better understand sedimentation phenomena and contribute to the broader field of soft matter physics.

9 Summary [L. Annuss] and [R.-M. Lehner]

This study aimed to explore three different aspects of the hard sphere systems the crystallization process and its description through order parameters, the Kosterlitz-Thouless phase transition in two dimensions and the sedimentation process in an external gravitational field. To simulate the hard sphere system, a Monte-Carlo Markov-Chain algorithm was illustrated and implemented. This algorithm was tested against theoretical prediction. The pair correlation function $g(r)$ and the pressure p were selected as comparison parameters. $g(r)$ showed good agreement with the theoretical prediction in the gas, liquid and crystalline phase. The pressure also fits remarkably well with the Carnahan-Starling equation, which predicts the equation of state in the gaseous phase. It was not possible to compare the implementation to theoretical predictions for higher pressures because the implementation was too slow and took impractically long to equilibrate for dense systems. Because of this an existing implementation of the same algorithm was used to analyze the crystallization process and the Kosterlitz-Thouless phase transition.

The exploration of the crystallization process was classified with the Q_6 , q_6 , q_4 and q_6q_6 order parameters. An explanation of the drawback of using these parameters separately were given and how to reduce these limitations by combining them. With these parameters, it was shown that the hard disk system crystallizes into a crystal with a FCC structure together with single dispersed HCP layers.

First, the two-dimensional hard disk system was investigated by measuring an equation of state. Then the occurring phase were classified by their orientational and positional order. Here a liquid phase for $\eta < 0.701$, a hexatic phase for $0.702 < \eta < 0.718$ and lastly, a solid phase for $\eta > 0.719$ was identified. It was determined that the transition between hexatic and liquid phase is a first-order and between hexatic and solid a Kosterlitz-Thouless transition. The first-order transition was identified by the typical coexistence of the two participating phases. and the Kosterlitz-Thouless by the creation of bounded dislocations and their later unbinding.

In the context of sedimentation, the hard sphere system was examined under the influence of an external gravitational field. By introducing gravity and implementing fixed boundaries in the vertical direction, the simulation revealed the formation of a solid phase in the lower layers and a phase transition occurring in the middle layers. The height-dependent packing fraction $\eta(z)$ and pressure $P(z)$ were measured, showing reasonable agreement with the Carnahan-Starling equation of state. Although the pressure was slightly underestimated, likely due to limited equilibration time, the results demonstrated how gravity induces a density gradient leading to phase coexistence. Structural analysis identified the development of FCC and HCP crystal structures within the sedimented layers. This investigation highlighted the complex behaviors that arise when hard sphere systems are subjected to gravitational forces, offering insights into sedimentation dynamics and phase transitions that extend the understanding of such systems.

Ultimately, the hard disk system proves to be far more fascinating than its simplicity suggests, revealing a wealth of complex behaviors and insights that extend well beyond initial expectations

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10 Appendix

10.1 Simulation for $\eta = 0.6$

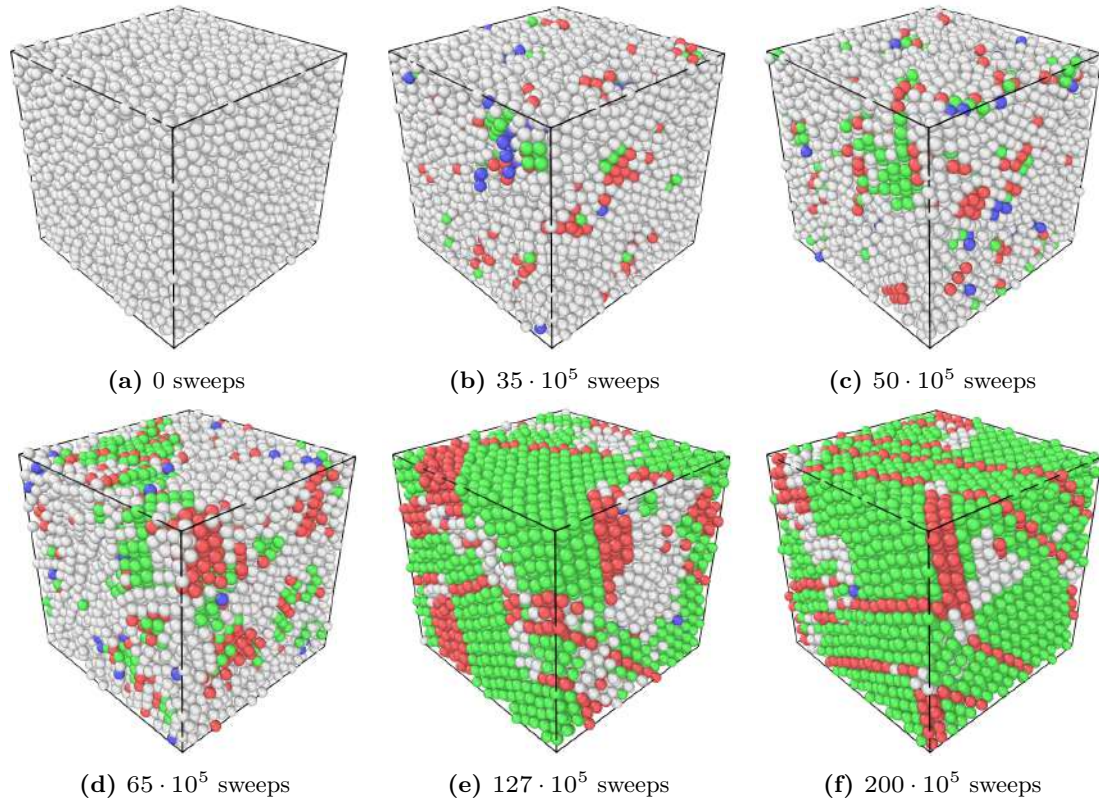


Figure 49: States of the simulation for $\eta = 0.6$ at different times.

10.2 Simulation for $\eta = 0.54$

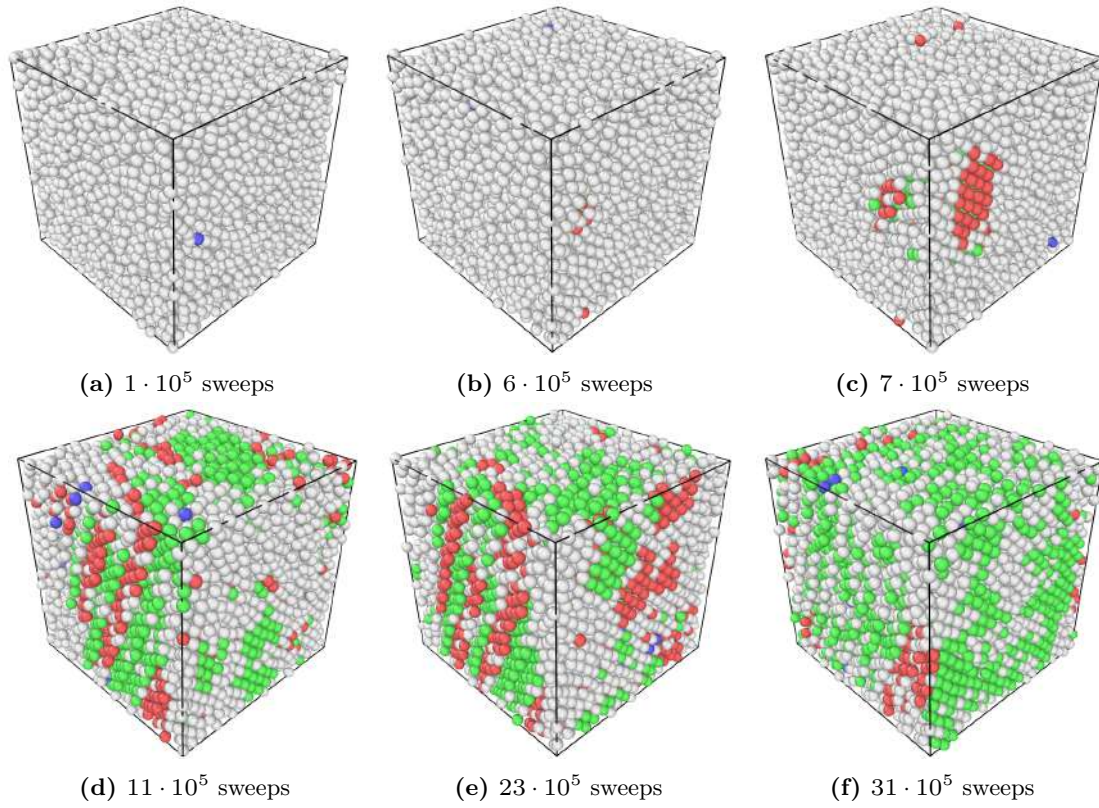


Figure 50: States of the simulation for $\eta = 0.54$ at different times.

10.3 Melting for $\eta = 0.49$

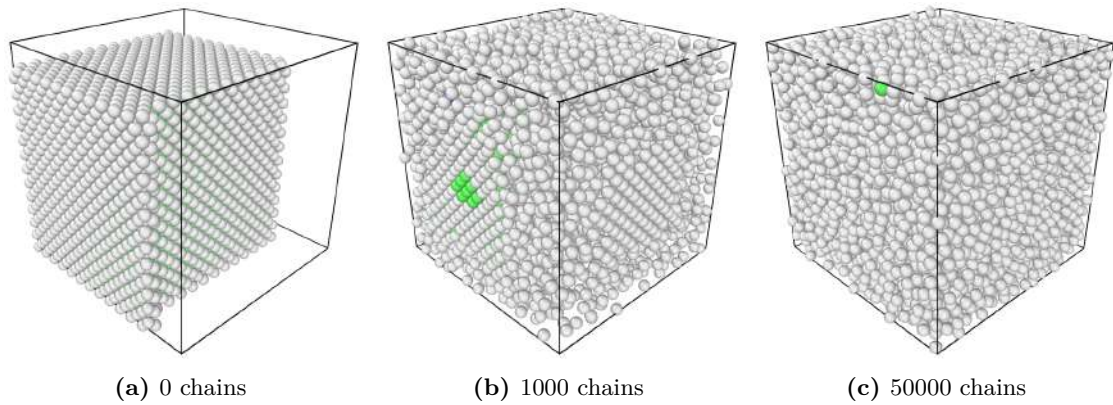


Figure 51: States of the simulation of the melting process at $\eta = 0.49$ in three dimensions.